



UNIVERSITI SAINS MALAYSIA

School
of
Electrical and Electronic Engineering

EEM 441

**Control, Instrumentation
& FPGA Laboratory**

Laboratory Manual

Experiments 1–10

Contents

Experiment 1: DC Motor / PID Control — Speed	2
Experiment 2: DC Motor / PID Control — Position.....	8
Experiment 3: Introduction to Altera Quartus II Software, and Interfacing FPGA to DIP Switch and Seven Segment	15
Experiment 4: LCD Interfacing to Altera DE2 Board, and Stepper Motor Control with FPGA	25
Experiment 5: Interfacing FPGA to Analog-to-Digital Converter.....	42
Experiment 6: Interfacing FPGA to Digital-to-Analog Converter.....	47
Experiment 7: Computer-Based Motor Speed Measurement.....	51
Experiment 8: Temperature Control Using Raspberry Pi.....	58
Experiment 9: IIoT Monitoring & Alarm Visualization for a PLC-Controlled Machine	63
Experiment 9: IIoT Monitoring & Alarm Visualization for a PLC-Controlled Machine	75





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Universiti Sains Malaysia, Engineering Campus

SEMESTER 1, 2026/2027

EEM 441

Control & Instrumentation Laboratory

Experiment 1

DC MOTOR / PID CONTROL — SPEED

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 1

DC Motor / PID Control — Speed

Objectives

1. To study the open-loop speed control of a DC motor.
2. To study closed-loop speed control using P, PD, PI, and PID controllers.

Equipment / Software Required

- Controller kit
- Cathode ray oscilloscope
- BNC connectors with leads
- Multimeter

Procedure

(a) Open-loop control of DC motor

1. Switch on the supply and measure all constant voltages; calibrate variable voltages in terms of voltage and angle (degrees).
2. Disconnect all cable wires from the hardware module (Figure 1.1).
3. Connect *Output A* of the motor plant to Channel 2 of the oscilloscope.
4. Connect *Adjust A* of the motor plant to *Input A* (speed control of the plant) and to Channel 1 of the oscilloscope (only positive voltages are considered).
5. Turn ON the power switch.
6. Slowly turn potentiometer-A clockwise (0 to +50) and anti-clockwise (0 to -50). Observe and describe the motor operation.
7. Disconnect *Adjust A* and connect the motor *Input A* to *Adjust B* (only positive voltages are considered).
8. Vary supply B from 30° to 270° in steps of 30° and study the variation in reference speed (voltage) and output speed (voltage). Plot the error curve.
9. Comment on your observations and results.

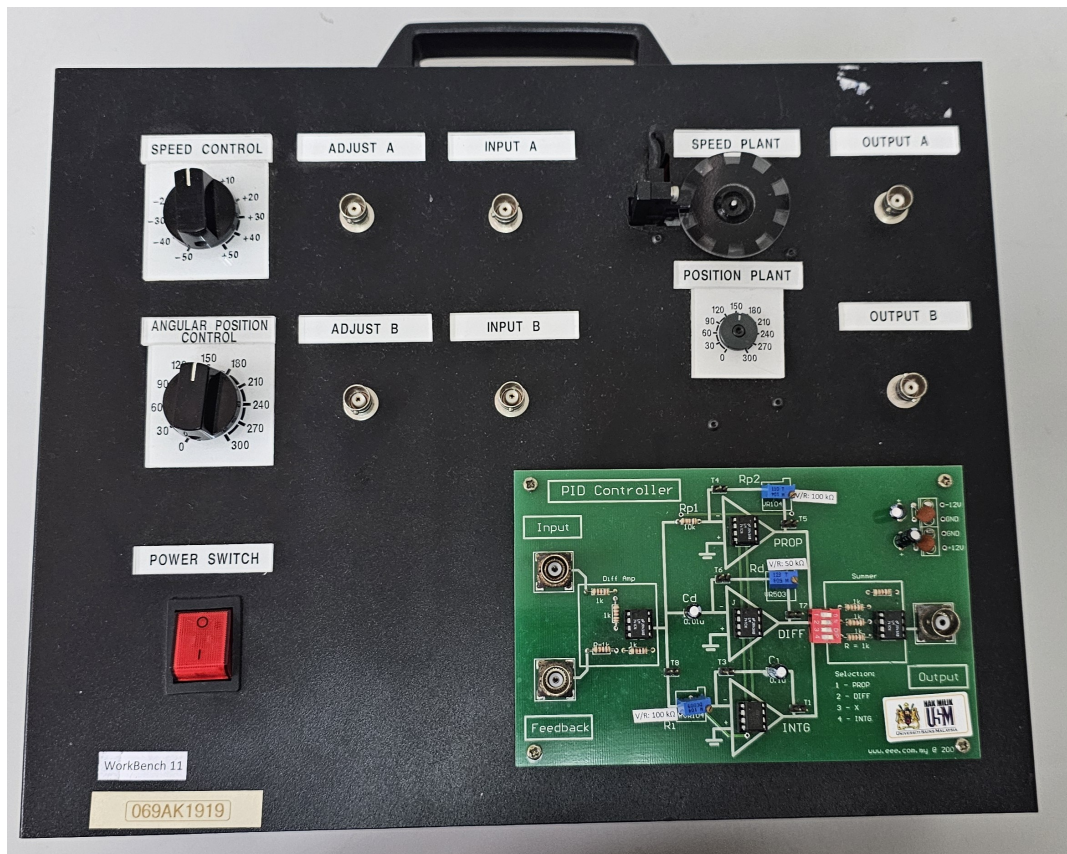


Figure 1.1: DC Motor / PID Control trainer.

Table 1.1: Open-loop speed response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
+270				
+240				
+210				
+180				
+150				
+120				
+90				
+60				
+30				

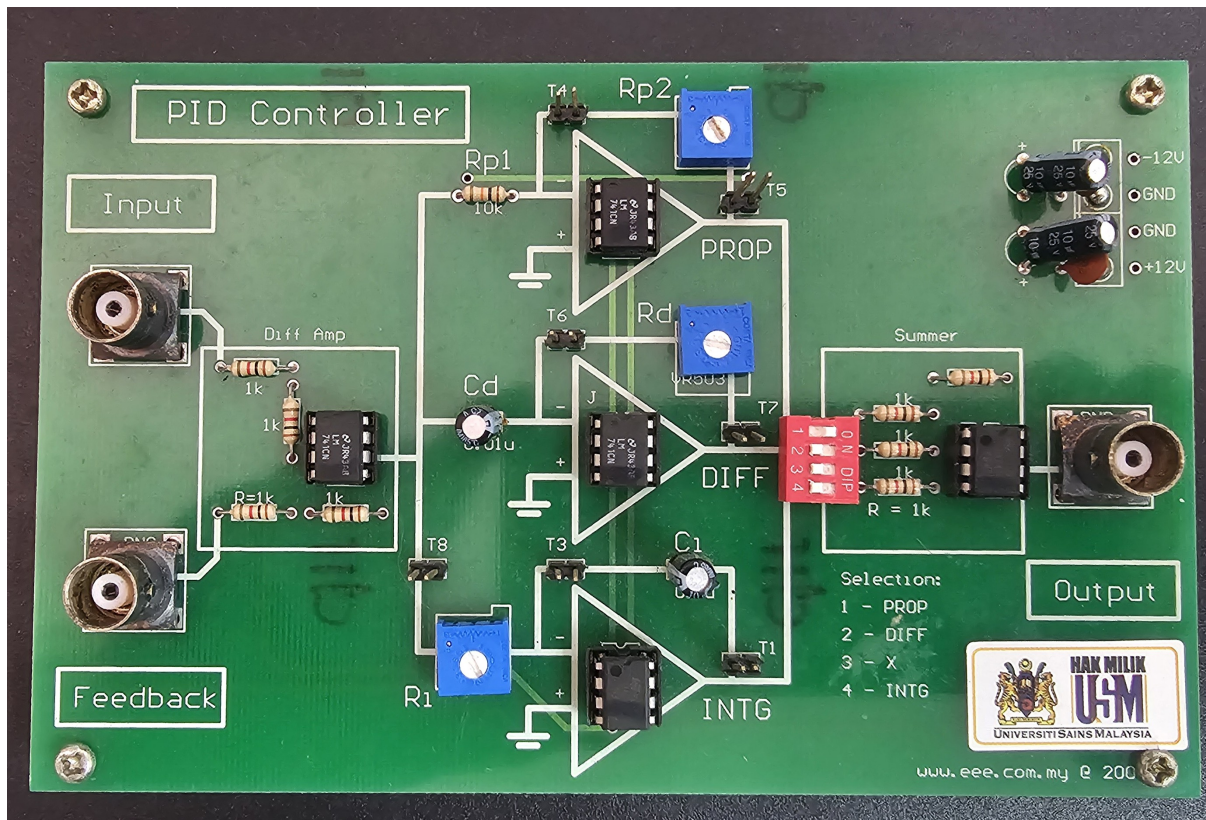


Figure 1.2: PID Control module.

(b) Closed-loop control of speed of DC motor

(i) Using P-controller with feedback through optical encoder

1. Select the P-controller by setting PROP on the controller (DIP switch 1) to ON (refer to Figure 1.2).
2. Connect *Adjust B* of the plant to the input of the PID controller.
3. Connect *Output A* of the plant to the controller feedback.
4. Connect the controller output to *Input A* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. +180).
6. Adjust trimmer R_{p2} (varies K_p) to obtain an optimum response (minimum steady-state error).
7. Turn potentiometer-B to vary the speed of the motor. Record the input and output voltages for various positions in tabular form.

(ii) Speed control with PI-controller

1. On the PID controller card, set trimmers R_p and R_d to minimum (counter-clockwise) while R_i is set to maximum (clockwise).
2. Select the PI-controller by setting PROP and INTG to ON.
3. Slowly vary trimmer R_i to obtain an optimum output response (minimum steady-state error). If gain is insufficient (or overshoots), adjust R_{p2} .

4. Turn potentiometer-B to vary the speed of the motor. Record the input and output voltages for various positions (Table 1.2). Repeat step 3 if necessary to fine-tune the response.
5. Compare the experimental results for the P-controller and the PI-controller.

(iii) Speed control with PD-controller

1. Switch on the supply and measure all constant voltages; calibrate variable voltages in terms of voltage and angle.
2. On the PID controller card, set trimmers R_p and R_d to minimum (counter-clockwise) while R_i is set to maximum (clockwise).
3. Select the PD-controller by setting PROP and DIFF to ON; all other switches OFF.
4. Vary supply B from 30° to 270° in steps of 30° and study the variation in reference speed (V_{in}) and output speed (V_o). Plot the error curve.
5. Study the effect of varying R_d on the output.

(iv) Speed control with PID-controller

1. Adjust potentiometer-B to 2 V (around $+180^\circ$).
2. Set trimmer R_{p2} to minimum to reduce the damping ratio.
3. Set trimmer R_i to minimum. The output should start to vibrate.
4. Slightly increase R_i until the system starts oscillating between two points (pure oscillation).
5. Slightly increase R_{p2} to bring the system to a damped condition; it should oscillate for a few cycles.
6. Apply a small force to hold the motor (the oscilloscope will show the motor output drop to 0 V); observe and describe the response when the force is released.
7. Select the PID-controller by setting PROP, INTG, and DIFF to ON.
8. Slowly vary R_d to obtain the optimum response.
9. Compare its operation with the P, PI, and PD controllers under disturbance.
10. As before, take observations by varying the applied voltage.

Table 1.2: Closed-loop speed response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
+270				
+240				
+210				
+180				
+150				
+120				
+90				
+60				
+30				

Report

1. Compare the performance of P, PI, PD, and PID controllers.
2. What will happen if all types of controllers are connected in series?
3. What are the different types of encoders? Explain the working principle of an optical encoder with the help of suitable diagrams.



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Experiment 2

DC MOTOR / PID CONTROL — POSITION

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 2

DC Motor / PID Control – Position

Objectives

1. To study open-loop position control of a DC motor.
2. To study feedback (closed-loop) control using P, PI, PD, and PID controllers.
3. To implement position control of a DC motor using a potentiometer and a resistance-to-voltage converter.

Equipment / Software Required

- Controller kit
- Cathode ray oscilloscope
- BNC connectors with leads
- Multimeter

Procedure

(a) Position control without feedback

1. Connect *Output B* of the motor plant to Channel 2 of the oscilloscope.
2. Connect *Adjust A* of the motor plant to *Input B* (position control) of the plant and to Channel 1 of the oscilloscope.
3. Turn on the power switch.
4. Slowly turn potentiometer-A clockwise (0 to +50) and anti-clockwise (0 to –50). Observe and describe the motor operation. Can you control the angular position of the motor?
5. Turn off the motor by disconnecting *Input B*.

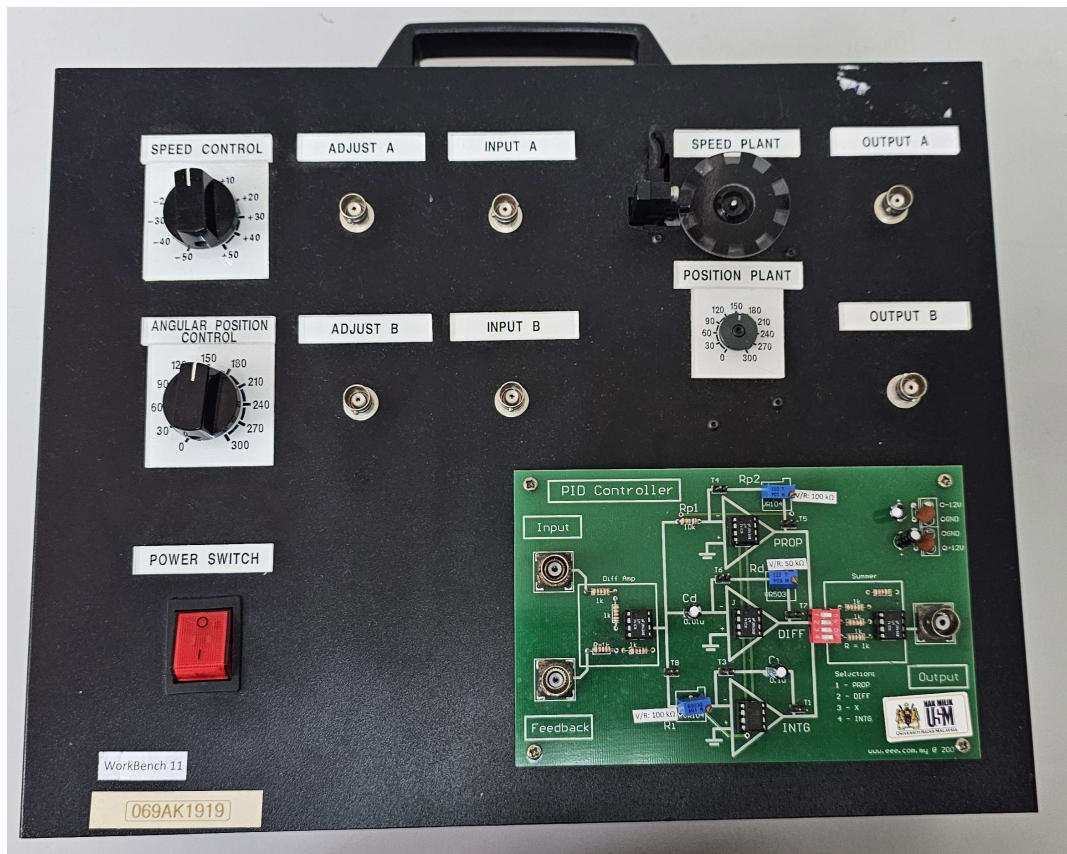


Figure 2.1: DC Motor / PID Control trainer.

(b) Closed-loop position control

(i) Position control with proportional feedback control

The motion of a motor can be converted into rotational position with the help of a suitable gearbox. The electrical signal, proportional to (angular) position, is obtained with the help of a circular potentiometer of good accuracy.

This signal is fed back to the summer to extract an error signal, which is fed to the controller to obtain optimum performance. Different controllers may be tried for this case, to study them: use the PID Controller daughterboard shown in Figure 2.2, selecting the desired stage(s) via its DIP switch.

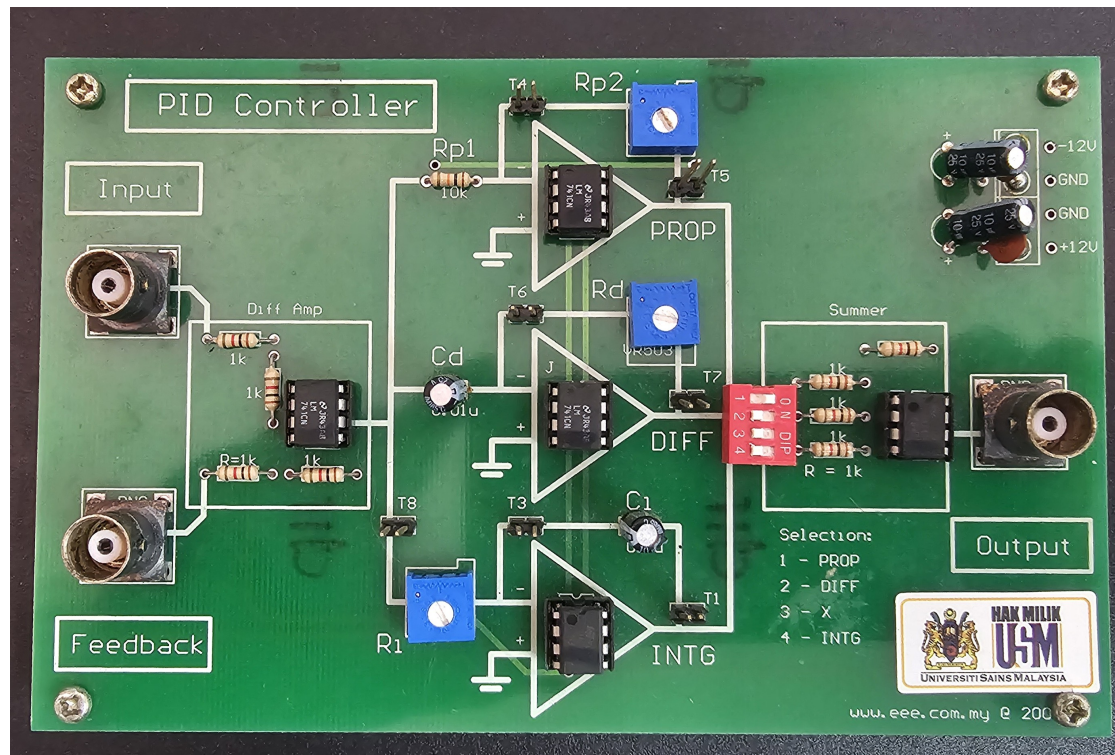


Figure 2.2: PID Control module.

1. Select the P-controller by setting PROP on the controller (DIP switch 1) to ON; all other switches OFF.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. +180).
6. Adjust trimmer R_{p2} (varying K_p) to obtain an optimum response (minimum steady-state error).
7. Turn potentiometer-B to vary the position of the motor. Can the position be controlled in this case?
8. Record the input and output voltages for various positions in Table 2.1.

Table 2.1: Position control response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
+270				
+240				
+210				
+180				
+150				
+120				
+90				
+60				
+30				

(ii) Position control with feedback and PI-controller

1. Select the PI-controller by setting PROP and INTG to ON.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. +180).
6. Slowly vary the respective trimmers R_{p2} and R_i to obtain an optimum output response.
7. Turn potentiometer-B to vary the position of the motor. Record the input and output voltages for various positions.

(iii) Position control with PD-controller

1. Select the PD-controller by setting PROP and DIFF to ON; all other switches OFF.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. +180).
6. Slowly vary the respective trimmers R_{p2} and R_d to obtain an optimum output response.
7. Turn potentiometer-B to vary the position of the motor. Record the input and output voltages for various positions.

(iv) Closed-loop control of position of motor using PID controller

1. Select the PID-controller by setting PROP, INTG, and DIFF to ON.
2. Connect *Adjust B* of the plant to the *input* of the PID controller.
3. Connect *Output B* of the plant to *Feedback* of the controller.
4. Connect the controller *Output* to *Input B* of the plant.
5. Turn potentiometer-B of the motor plant to a middle position (e.g. +180).
6. Adjust the value of trimmers R_{p2} , R_i , and R_d to obtain an optimum response (minimum steady-state error, quick operation, without overshoot or oscillations).
7. Turn potentiometer-B to vary the position of the motor.
8. Record the input and output voltages for various positions in Table 2.2.

Table 2.2: Closed-loop PID position response.

Position	V_{in} (Channel I)	V_{out} (Channel II)	Error ($V_{out} - V_{in}$)	Remarks
+270				
+240				
+210				
+180				
+150				
+120				
+90				
+60				
+30				

Report

1. Briefly explain the working of circuits used for the conversion of resistance to voltage.
2. Describe in detail the applications of position control.



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Experiment 3

FPGA — INTRODUCTION TO ALTERA QUARTUS II
SOFTWARE, AND INTERFACING FPGA TO DIP SWITCH AND
SEVEN SEGMENT

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 3

Introduction to Altera Quartus II Software, and Interfacing FPGA to DIP Switch and Seven Segment

Objectives

1. To use schematic entry as a design method in Altera Quartus II.
2. To use Verilog as a design entry method in Altera Quartus II.
3. To combine schematic and Verilog as a single design entry in Altera Quartus II.
4. To simulate the circuit using Altera Quartus II.
5. To download the design onto an Altera DE2 board.
6. To interface a DIP switch and a seven-segment display to the FPGA.

Equipment / Software Required

1. Altera Quartus II software
 - version 9/10/12 for Cyclone II
 - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Function generator

Introduction

Seven-segment displays are widely used to show numbers. The DE2 board provides eight seven-segment displays, 18 DIP switches, four push-button switches, and 16 LEDs. These components are hardwired on the board and are each assigned to specific FPGA pins — refer to the DE2 user manual for the full pin map.

Procedure

Activity 1 — Schematic entry

Objective: use schematic entry to design a 4-to-1 multiplexer in Quartus II.

1. Open Quartus II and start the New Project Wizard.
 - Working directory: fpga; project name: activity1; top-level entity: mux_sch.
 - Device family: Cyclone II (EP2C35F672C6N) or Cyclone IV (EP4CE115F29C7N).

- Accept the default EDA tool settings (synthesis, simulation, timing analysis) and finish.
2. Create a new Block Diagram/Schematic file.
 3. Place three 3-input AND gates and one 4-input OR gate from the primitives library, and wire them as shown in Figure 3.1 using rubberbanding to connect components.
 4. Add and name the input pins (i0–i3, s0, s1) and the output pin (out).
 5. Save as mux_sch.bdf and start compilation.
 6. Create a Vector Waveform File to simulate the design: insert input nodes i0–i3, s0, s1, and output node out.
 7. Assign logic values to the inputs, save as mux_sch.vwf, then run Compilation and Simulation.
 8. Record the outputs, then repeat for each input combination in Table 3.1, saving the waveform file each time.
 9. Print and submit the schematic diagram and simulation result.

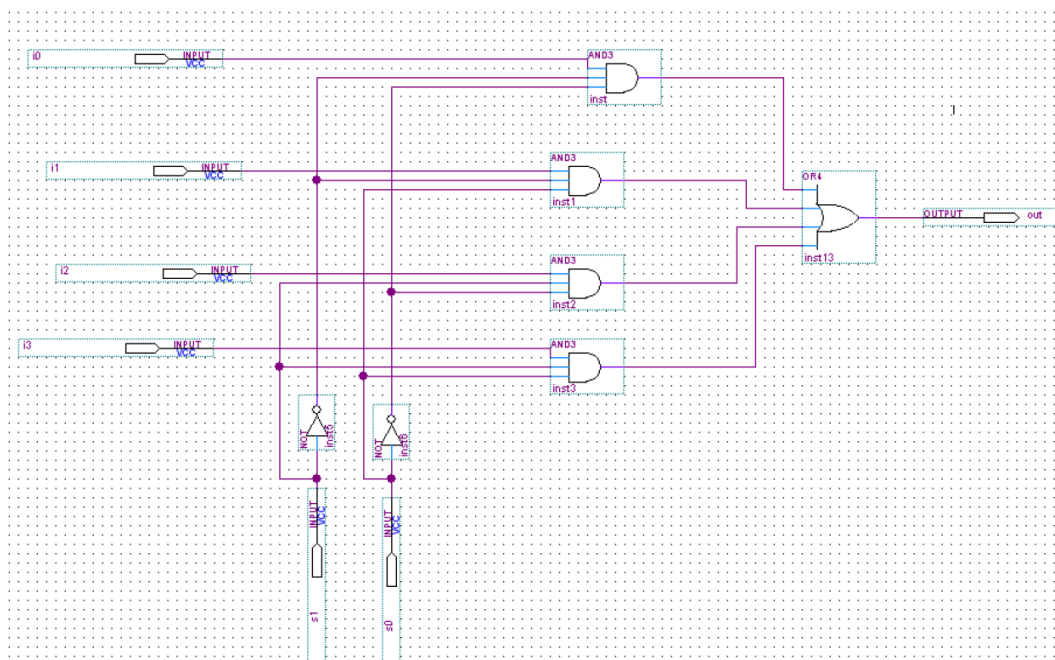


Figure 3.1: Mux Schematic.

Table 3.1: Input combinations to simulate.

S0	S1	I0	I1	I2	I3
0	0	1	1	0	0
0	1	0	1	1	0
1	0	1	1	0	1
1	1	1	1	0	0

Activity 2 – Verilog entry

Objective: use Verilog code as a design entry method in Quartus II.

1. Start a new project (Activity2, top level mux_ver) with the same device and EDA tool settings as Activity 1.
2. Create a new Verilog file and enter the code in Listing 3.1.
3. Save as mux_ver, then compile the design.
4. Create a Vector Waveform File; insert input nodes i0–i3, s0, s1, and output node out.
5. Assign logic values matching Table 3.1, save as mux_ver.vwf, then start simulation.
6. Inspect the result in the Simulation Report – Simulation Waveform.
7. Change the input values (saving before each run) and re-simulate.
8. View the synthesized schematic via **Tools** → **Netlist Viewer** → **RTL Viewer**, shown in Figure 3.2.
9. Print and submit the Verilog code and simulation result.

```
mux_ver.v
1  module mux_ver(out, i0, i1, i2, i3, s1, s0);
2      output out;
3      input i0, i1, i2, i3;
4      input s1, s0;
5      reg out;
6
7      always @(s1 or s0 or i0 or i1 or i2 or i3)
8      begin
9          case ({s1, s0})
10             2'b00: out = i0;
11             2'b01: out = i1;
12             2'b10: out = i2;
13             2'b11: out = i3;
14             default: out = 1'bx;
15          endcase
16      end
17  endmodule
```

Listing 3.1: mux_ver.v — Verilog 4-to-1 multiplexer.

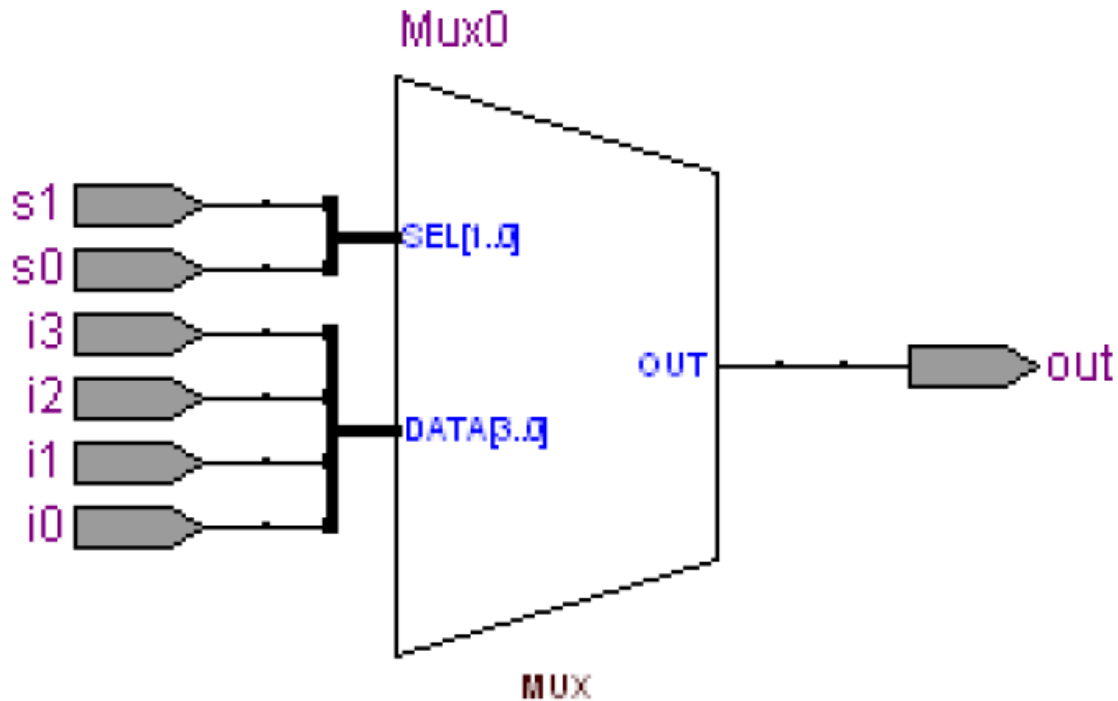


Figure 3.2: RTL Viewer output for mux_ver (Mux0).

Activity 3 – Combining schematic and Verilog entry

Objective: combine a Verilog module with schematic entry in a single design.

1. Start a new project (activity3, top level mux_schver) with the same device and EDA tool settings.
2. Create a new Verilog HDL file with the code in Listing 3.2; save as mux_vr1.
3. Generate a reusable symbol from the file via **File** → **Create/Update** → **Create Symbol Files for Current File**. The resulting block is shown in Figure 3.3.
4. Create a new Block Diagram/Schematic file, place the mux_vr1 symbol from the project library, and connect its inputs and output to input/output pads as shown in Figure 3.4.
5. Save as mux_schver.bdf and start compilation.
6. Create a Vector Waveform File, insert the same nodes as before, assign logic values, save as mux_schver.vwf, and run Compilation and Simulation.
7. Record the outputs for the combinations in Table 3.1, saving the waveform file each time.
8. Print and submit the schematic diagram, Verilog code, and simulation result.

```

mux_vr1.v
1  module mux_vr1(out, i0, i1, i2, i3, s1, s0);
2      output out;
3      input i0, i1, i2, i3;
4      input s1, s0;
5      reg out;
6
7      always @(s1 or s0 or i0 or i1 or i2 or i3)
8          begin
9              case ({s1, s0})
10                 2'b00: out = i0;
11                 2'b01: out = i1;
12                 2'b10: out = i2;
13                 2'b11: out = i3;
14                 default: out = 1'bx;
15             endcase
16         end
17     endmodule

```

Listing 3.2: mux_vr1.v — Verilog module to be reused as a schematic symbol.

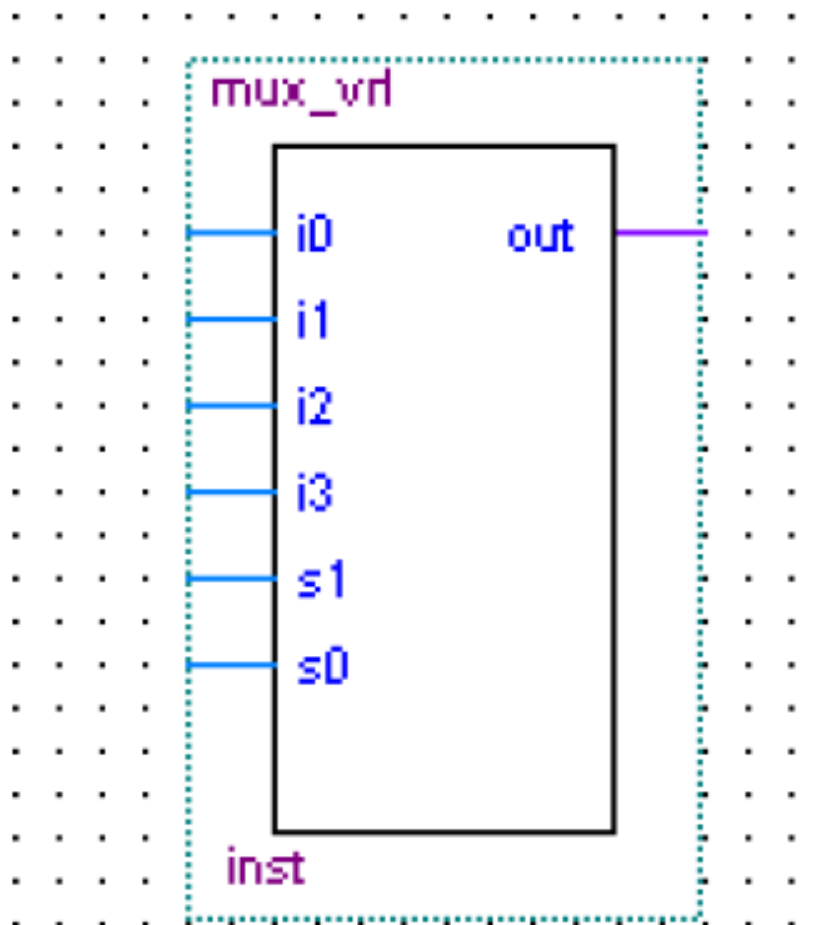


Figure 3.3: Auto-generated mux_vr1 symbol block.

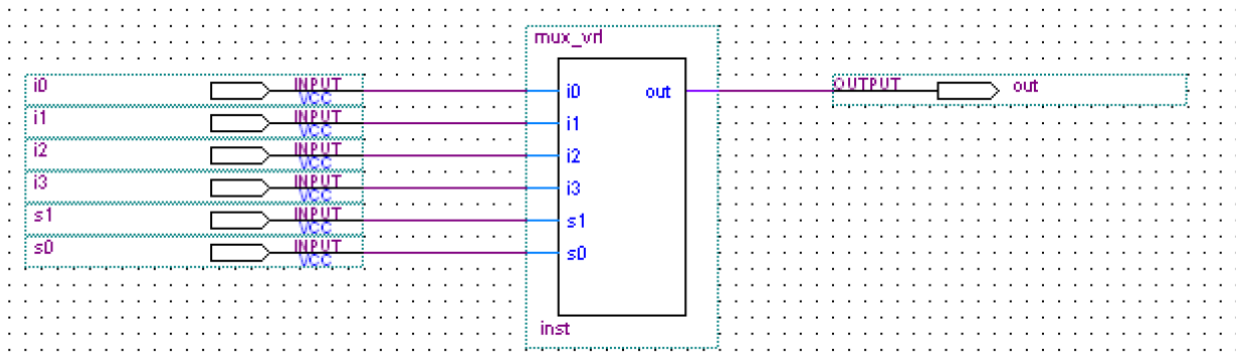


Figure 3.4: Combined schematic mux_schver.

Activity 4 – Timing analysis

Objective: analyse the maximum operating speed of a design.

1. Start a new project (Activity4, top level counter) with the same device and EDA tool settings.
2. Create a new Verilog file with the code in Listing 3.3; save as counter and compile.
3. Create a Vector Waveform File; insert cclock and cclear as inputs and Q as output.
4. Force cclear low, assign a clock signal to cclock, save as counter.vwf, then start simulation.
5. Inspect the result in the Simulation Report – Simulation Waveform.
6. Run **Processing** → **Start** → **Start Classic Timing Analyzer**.
7. Read the maximum clock frequency from the Timing Analyzer Summary (Figure 3.5).
8. Print and submit the Verilog code and simulation result.

counter.v

```

1  module counter(Q, clock, clear);
2      output [3:0] Q;
3      input clock, clear;
4      reg [3:0] Q;
5
6      always @(posedge clear or negedge clock)
7      begin
8          if (clear)
9              Q = 4'd0;
10         else
11             Q = (Q + 1) % 16;
12     end
13 endmodule

```

Listing 3.3: counter.v – 4-bit up counter with asynchronous clear.

Timing Analyzer Summary									
	Type	Slack	Required Time	Actual Time	From	To	From Clock	To Clock	Failed Paths
1	Worst-case tco	N/A	None	7.410 ns	Q[3]~reg0	Q[3]	clock	--	0
2	Clock Setup: 'clock'	N/A	None	Restricted to 233.64 MHz (period = 4.280 ns)	Q[0]~reg0	Q[3]~reg0	clock	clock	0
3	Total number of failed paths								0

Figure 3.5: Timing analyzer summary for the counter design.

Activity 5 – Downloading to the DE2 board

Objective: download a Verilog design onto the Altera DE2 board and drive the seven-segment display.

1. Start a new project (activity5, top level counta) with the same device and EDA tool settings.
2. Create a new Verilog HDL file with the code in Listing 3.4; save as counta.
3. Compile and simulate the design.
4. Assign each output pin to a physical device pin using **Assignment** → **Pin Planner**, per Table 3.2 (refer to the DE2 user manual for pin locations).
5. Save, then recompile the project.
6. Download the configuration to the DE2 board via **Tools** → **Programmer** → **Hardware Setup** → USB-Blaster → **Start**.
7. Record the pattern displayed on the seven-segment display.
8. Change the assignment to assign `q[6:0] = 7'b1111000;`, save, recompile, and download again. Record the new displayed pattern.

```
counta.v
1 module counta(q);
2     output [6:0] q;
3     assign q[6:0] = 7'b0010000;
4 endmodule
```

Listing 3.4: counta.v – static seven-segment pattern.

Table 3.2: Seven-segment output pin assignment (DE2 board).

#	Node name	Direction	Pin
1	Q[6]	Output	
2	Q[5]	Output	
3	Q[4]	Output	
4	Q[3]	Output	
5	Q[2]	Output	
6	Q[1]	Output	
7	Q[0]	Output	

Activity 6 – 16-bit binary counter with frequency divider

Objective: design a 16-bit binary counter driven through a frequency divider.

1. Open Quartus II and write the frequency-divider module in Listing 3.5 (seven), which lowers the incoming clock frequency (Device family: Cyclone II (EP2C35F672C6N) or Cyclone IV (EP4CE115F29C7N)).
2. Create a reusable symbol for seven.
3. Write the 4-bit counter module in Listing 3.6 (led).
4. Create a reusable symbol for led.
5. Draw the schematic in Figure 3.6, connecting the seven block's most significant output bit q[16] to the clock input of the led block.
6. Assign pin numbers as shown in Table 3.3 (refer to the DE2 user manual).
7. Use a function generator to supply the clock signal, using 1 MHz as the clock input.
8. Determine the effective input frequency reaching the led module.
9. Discuss your result based on the observed LED output display.

seven.v – frequency divider

```
1 module seven(q, clk);
2   output [15:0] q;
3   input clk;
4   reg [15:0] q;
5
6   always @(posedge clk)
7   begin
8     q = (q + 1) % 65536;
9   end
10 endmodule
```

Listing 3.5: seven.v – 16-bit frequency divider.

led.v – 4-bit counter

```
1 module led(x, clk);
2   input clk;
3   output [3:0] x;
4   reg [3:0] x;
5
6   always @(posedge clk)
7   begin
8     x = (x + 1) % 16;
9   end
10 endmodule
```

Listing 3.6: led.v – 4-bit binary counter.

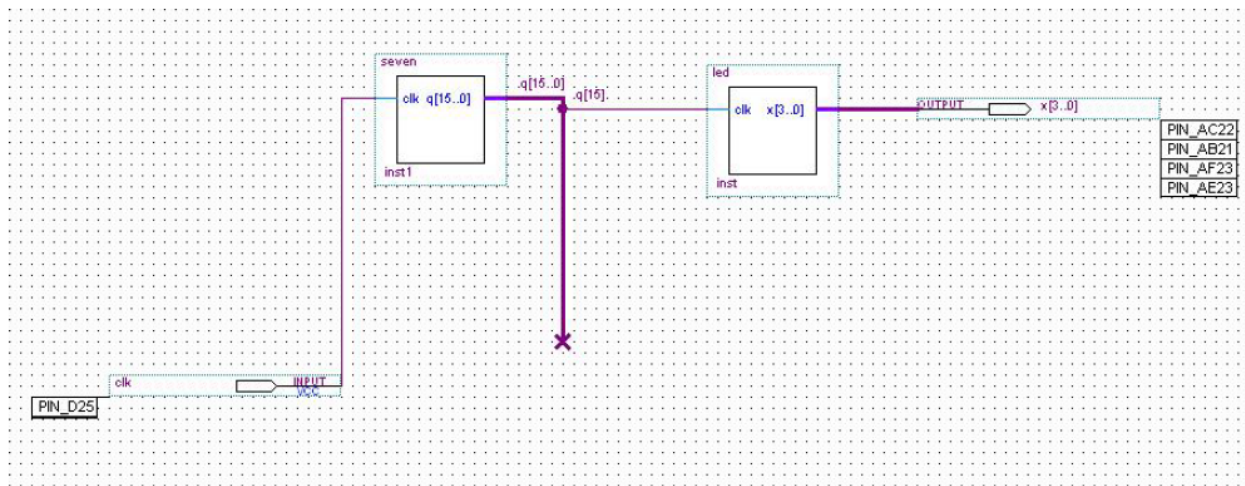


Figure 3.6: Schematic diagram for the 16-bit binary counter with frequency divider.

Table 3.3: Pin assignment for the 4-bit binary counter.

#	Node name	Direction	Pin location
1	clk	Input	
2	X[3]	Output	
3	X[2]	Output	
4	X[1]	Output	
5	X[0]	Output	

Activity 7 – 0–99 counter on seven-segment display

Objectives: (1) design a counter that counts 0 to 99; (2) display the result on the seven-segment displays.

1. Using a DIP switch as the trigger input, write Verilog code for a counter that counts from 0 to 99 once the switch is triggered.
2. Decode and display the result (tens and units digits) on two seven-segment displays.
3. Print and submit the Verilog code and simulation result.

Activity 8 – Running light

Objective: design a running-light display on the Altera DE2 board.

1. Using a DIP switch as the input, write Verilog code to implement a running-light (chasing LED) pattern.
2. Use three DIP switches to select between three different running-light patterns.
3. Print and submit the Verilog code and simulation result.



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SEMESTER 1, 2026/2027

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Experiment 4

LCD INTERFACING TO ALTERA DE2 BOARD, AND STEPPER
MOTOR CONTROL WITH FPGA

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 4

LCD Interfacing to Altera DE2 Board, and Stepper Motor Control with FPGA

Objectives

1. To use Verilog code as a design entry method in Altera Quartus II Software.
2. To interface an LCD module to the Altera DE2 board.
3. To design digital circuits in an FPGA to control a stepper motor.
4. To simulate the circuit using Altera Quartus II Software.
5. To download the design onto an Altera DE2 board.

Equipment / Software Required

1. Altera Quartus II software
 - version 9/10/12 for Cyclone II
 - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Parallax 4-phase, 12 V, 75 Ω unipolar stepper motor
4. ULN2003 Darlington transistor array (motor driver)

Introduction

In recent years, LCDs have largely replaced simple LED displays: they can show numbers, characters, and graphics, and are relatively easy to program. Table 4.1 lists the LCD pins relevant to programming.

The *enable* (E) pin is used by the LCD to latch the information presented on its data pins: a high-to-low transition on E must be applied for the LCD to latch the data, and this pulse must be at least 450 ns wide. The required timing is shown in Figure 4.1.

Two important internal registers are selected via the RS pin. When RS = 0, the *instruction/command* register is selected, allowing commands to be sent to the LCD (Table 4.2 lists the available command codes). When RS = 1, the *data* register is selected, allowing ASCII character data to be written for display, sent through data pins D0–D7.

It is good practice to check the busy flag before writing to the LCD: with RS = 0 and $\overline{R/\overline{W}} = 1$, the busy flag appears on D7. While D7 = 1 the LCD is busy with internal operations and

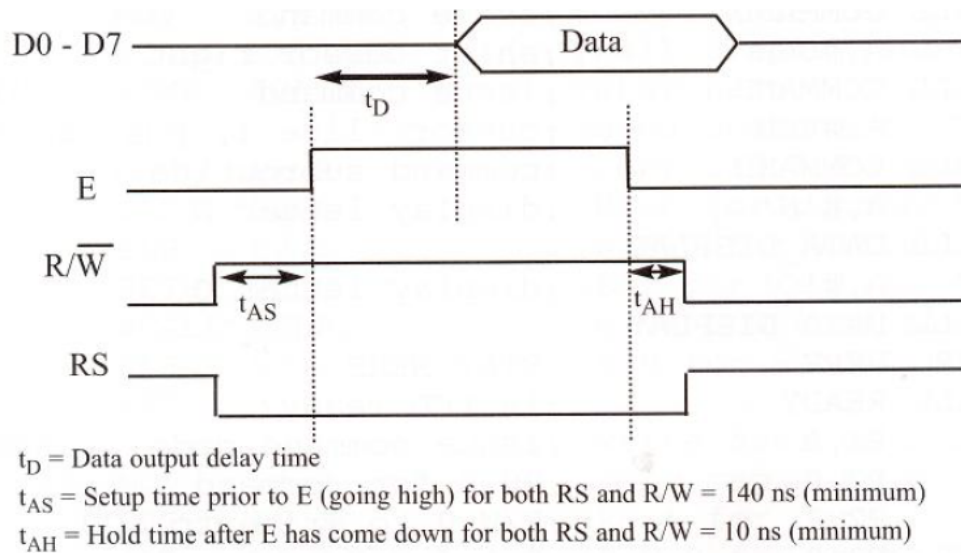
will not accept new data; once $D7 = 0$, the LCD is ready to receive the next byte.

Unlike an ordinary DC motor, a stepper motor does not spin freely when power is applied — it must be driven with a continuously pulsed sequence of specific coil-energising patterns. Each pulse advances the rotor by a fixed step angle (for this motor, 3.6° per step), which is what gives stepper motors their positioning precision: as long as speed and torque stay within the motor's limits, the controller always knows the rotor's exact position.

The rotor is driven by the interaction of magnetic fields as coils are switched on and off in sequence. Table 4.4 shows the normal four-coil unipolar step sequence; taking a step means driving two of the four coil-driver pins high at a time, in the order shown, through the ULN2003 driver.

Table 4.1: Pin description for the LCD module.

Pin	Symbol	I/O	Description
1	V_{ss}	—	Ground
2	V_{cc}	—	+5 V power supply
3	V_{ee}	—	Power supply to control contrast
4	RS	I	RS = 0: command register; RS = 1: data register
5	$R/\overline{W}$	I	0: write; 1: read
6	E	I/O	Enable (latches data on high-to-low transition)
7	DB0	I/O	8-bit data bus
8	DB1	I/O	
9	DB2	I/O	
10	DB3	I/O	
11	DB4	I/O	
12	DB5	I/O	
13	DB6	I/O	
14	DB7	I/O	



Note: Read requires an L-to-H pulse for the E pin.

Figure 4.1: LCD write-cycle timing diagram.

Table 4.2: LCD command codes.

Code (hex)	Command to LCD instruction register
1	Clear display screen
2	Return home
4	Decrement cursor (shift cursor left)
6	Increment cursor (shift cursor right)
5	Shift display right
7	Shift display left
8	Display off, cursor off
A	Display off, cursor on
C	Display on, cursor off
E	Display on, cursor blinking
F	Display on, cursor blinking
10	Shift cursor position left
14	Shift cursor position right
18	Shift entire display left
1C	Shift entire display right
80	Force cursor to beginning of line 1
C0	Force cursor to beginning of line 2
38	2 lines, 5 × 7 dot matrix

Procedure

Activity 1 — Displaying lookup-table data on the LCD

Objective: display fixed text, stored in a lookup table, on the LCD.

1. Open Quartus II (Device family: Cyclone II (EP2C35F672C6N) or Cyclone IV (EP4CE115F29C7N))
2. Write the Verilog modules given in Listing 4.1 (lcd_t) and Listing 4.2 (LCD_Controller) to send commands and data to the LCD.
3. lcd_t uses a lookup table (LUT) to store the sequence of commands and characters to send to the LCD.
4. Create a reusable symbol for each module, as in Experiment 3.
5. Draw the schematic in Figure 4.2, connecting lcd_t's outputs to LCD_Controller's inputs, and LCD_Controller's outputs to the LCD interface pins.
6. Assign pin numbers as given in the Altera DE2 board manual (Cyclone II or Cyclone IV).
7. Download the program to the Altera DE2 board.
8. Discuss the result based on your observation of the LCD.

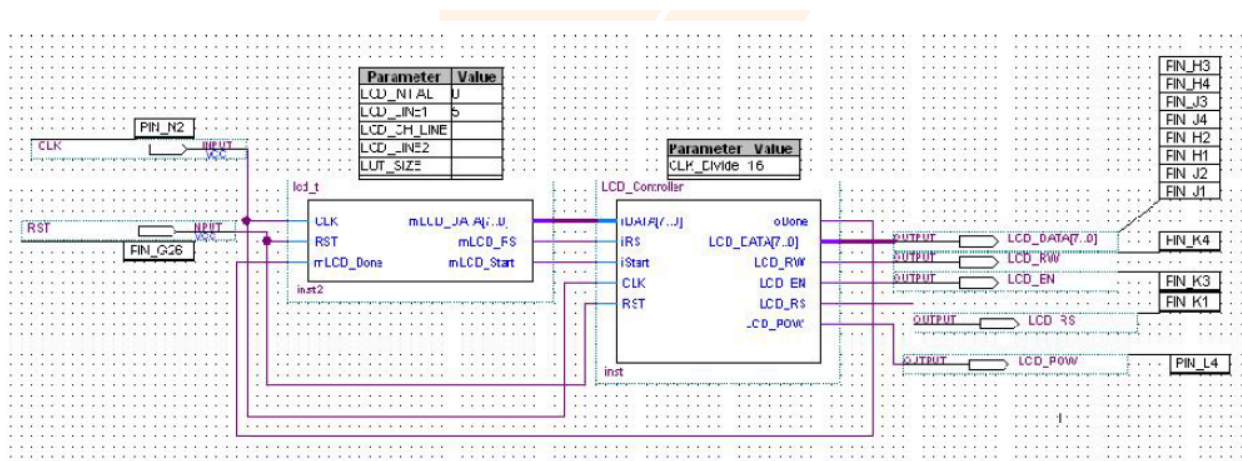


Figure 4.2: Schematic diagram for the LCD interface: lcd_t (lookup-table sequencer) feeding LCD_Controller (bus-timing controller).

Table 4.3: Pin assignment for the LCD interface.

#	Node name	Direction	Location
1	LCD_DATA[7]	Output	
2	LCD_DATA[6]	Output	
3	LCD_DATA[5]	Output	
4	LCD_DATA[4]	Output	
5	LCD_DATA[3]	Output	
6	LCD_DATA[2]	Output	
7	LCD_DATA[1]	Output	
8	LCD_DATA[0]	Output	
9	LCD_EN	Output	
10	LCD_POW	Output	
11	LCD_RS	Output	
12	LCD_RW	Output	
13	CLK	Input	
14	RST	Input	

lcd_t.v — Appendix 1: lookup-table sequencer (part 1 of 2)

```

1  module lcd_t (
2      // Host side
3      CLK, RST, mLCD_Done,
4      mLCD_DATA, mLCD_RS, mLCD_Start
5  );
6      // Host side
7      input  CLK, RST, mLCD_Done;
8      // Internal wires/registers
9      output [7:0] mLCD_DATA;
10     output mLCD_RS, mLCD_Start;
11     reg  [5:0] LUT_INDEX;
12     reg  [8:0] LUT_DATA;
13     reg  [5:0] mLCD_ST;
14     reg  [17:0] mDLY;
15     reg  mLCD_Start;
16     reg  [7:0] mLCD_DATA;
17     reg  mLCD_RS;
18     wire mLCD_Done;
19
20     parameter LCD_INITIAL = 0;
21     parameter LCD_LINE1   = 5;
22     parameter LCD_CH_LINE = LCD_LINE1 + 16;
23     parameter LCD_LINE2   = LCD_LINE1 + 16 + 1;
24     parameter LUT_SIZE    = LCD_LINE1 + 32 + 1;
25

```

```

26 // Sequencing state machine: send one LUT entry, wait for
27 // LCD_Controller to finish (mLCD_Done), pause, then advance.
28 always @(posedge CLK or negedge RST)
29 begin
30     if (!RST)
31     begin
32         LUT_INDEX <= 0;
33         mLCD_ST <= 0;
34         mDLY <= 0;
35         mLCD_Start <= 0;
36         mLCD_DATA <= 0;
37         mLCD_RS <= 0;
38     end
39     else if (LUT_INDEX < LUT_SIZE)
40     begin
41         case (mLCD_ST)
42         0: begin
43             mLCD_DATA <= LUT_DATA[7:0];
44             mLCD_RS <= LUT_DATA[8];
45             mLCD_Start <= 1;
46             mLCD_ST <= 1;
47         end
48         1: if (mLCD_Done) begin
49             mLCD_Start <= 0;
50             mLCD_ST <= 2;
51         end
52     end

```

lcd_t.v – Appendix 1: lookup-table sequencer (part 2 of 2, continued)

```

1         2: if (mDLY < 18'h3FFFE)
2             mDLY <= mDLY + 1;
3         else begin
4             mDLY <= 0;
5             mLCD_ST <= 3;
6         end
7         3: begin
8             LUT_INDEX <= LUT_INDEX + 1;
9             mLCD_ST <= 0;
10        end
11    endcase
12 end
13 end
14 // Lookup table: LCD_INITIAL entries are setup commands
15 // (function set, display on, clear, entry mode, home);
16 // LCD_LINE1 / LCD_LINE2 entries are ASCII characters for
17 // each line, with the MSB (bit 8) marking RS = 1 (data).
18 always
19 begin
20     case (LUT_INDEX)
21     LCD_INITIAL+0: LUT_DATA <= 9'h038; // function set
22     LCD_INITIAL+1: LUT_DATA <= 9'h00C; // display on

```

```

23     LCD_INITIAL+2: LUT_DATA <= 9'h001; // clear display
24     LCD_INITIAL+3: LUT_DATA <= 9'h006; // entry mode set
25     LCD_INITIAL+4: LUT_DATA <= 9'h080; // cursor home, line 1
26     // Line 1: "Welcome to the "
27     LCD_LINE1+0:   LUT_DATA <= 9'h120;
28     LCD_LINE1+1:   LUT_DATA <= 9'h157;
29     LCD_LINE1+2:   LUT_DATA <= 9'h165;
30     // ... remaining characters follow the same pattern ...
31     LCD_LINE1+15:  LUT_DATA <= 9'h120;
32     LCD_CH_LINE:   LUT_DATA <= 9'h0C0; // move to line 2
33     // Line 2: "    EEM 441 Lab "
34     LCD_LINE2+0:   LUT_DATA <= 9'h120;
35     LCD_LINE2+4:   LUT_DATA <= 9'h145;
36     // ... remaining characters follow the same pattern ...
37     LCD_LINE2+15:  LUT_DATA <= 9'h120;
38     default:       LUT_DATA <= 9'h000;
39   endcase
40   end
41   endmodule
42

```

Listing 4.1: lcd_t.v — lookup-table LCD message sequencer.

LCD_Controller.v — Appendix 2: bus-timing controller (part 1 of 2)

```

1  module LCD_Controller (
2      // Host side
3      iDATA, iRS, iStart, oDone, CLK, RST,
4      // LCD interface
5      LCD_DATA, LCD_RW, LCD_EN, LCD_RS, LCD_POW
6  );
7      parameter CLK_Divide = 16;
8
9      // Host side
10     input  [7:0] iDATA;
11     input  iRS, iStart;
12     input  CLK, RST;
13     output reg oDone;
14
15     // LCD interface
16     output [7:0] LCD_DATA;
17     output reg LCD_EN;
18     output LCD_RW;
19     output LCD_RS;
20     output LCD_POW;
21
22     // Internal registers
23     reg [4:0] Cont;
24     reg [1:0] ST;
25     reg preStart, mStart;
26
27     // Write-only interface: bypass iRS straight to LCD_RS

```

```

28 assign LCD_DATA = iDATA;
29 assign LCD_RW    = 1'b0;
30 assign LCD_RS    = iRS;
31 assign LCD_POW   = 1'b1;
32 always @(posedge CLK or negedge RST)
33 begin
34     if (!RST)
35     begin
36         oDone    <= 1'b0;
37         LCD_EN    <= 1'b0;
38         preStart <= 1'b0;
39         mStart    <= 1'b0;
40         Cont      <= 0;
41         ST        <= 0;
42     end
43     else
44     begin
45         // Rising-edge detect on iStart
46         preStart <= iStart;
47         if ({preStart, iStart} == 2'b01)
48         begin
49             mStart <= 1'b1;
50             oDone  <= 1'b0;
51         end

```

LCD_Controller.v – Appendix 2: bus-timing controller (part 2 of 2, continued)

```

1     if (mStart)
2     begin
3         case (ST)
4             0: ST <= 1; // wait setup
5             1: begin
6                 LCD_EN <= 1'b1; // raise enable
7                 ST <= 2;
8             end
9             2: if (Cont < CLK_Divide) // hold enable pulse width
10                Cont <= Cont + 1;
11            else
12                ST <= 3;
13            3: begin
14                LCD_EN <= 1'b0; // drop enable: latches data
15                mStart <= 1'b0;
16                oDone  <= 1'b1;
17                Cont   <= 0;
18                ST     <= 0;
19            end
20        endcase
21    end
22 end
23 end
24 endmodule

```

Listing 4.2: LCD_Controller.v – generates the LCD enable-pulse timing.

Activity 2 – Displaying DIP switch input on the LCD

Objective: display the live 8-bit DIP switch configuration on the LCD.

1. Write Verilog code that reads the 8-bit DIP switch configuration and displays it on the LCD (e.g. if switch 1 is turned on, the LCD should display 1 in the corresponding position).
2. Print and submit the Verilog code and simulation result.

Introduction

Unlike an ordinary DC motor, a stepper motor does not spin freely when power is applied – it must be driven with a continuously pulsed sequence of specific coil-energising patterns. Each pulse advances the rotor by a fixed step angle (for this motor, 3.6° per step), which is what gives stepper motors their positioning precision: as long as speed and torque stay within the motor's limits, the controller always knows the rotor's exact position.

The rotor is driven by the interaction of magnetic fields as coils are switched on and off in sequence. Table 4.4 shows the normal four-coil unipolar step sequence; taking a step means driving two of the four coil-driver pins high at a time, in the order shown, through the ULN2003 driver.

Table 4.4: Normal step sequence for a four-coil unipolar stepper motor.

Coil	Step 1	Step 2	Step 3	Step 4	Step 1
1 (B)	1	1	0	0	1
2 ($\bar{B}$)	0	0	1	1	0
3 (A)	1	0	0	1	1
4 ($\bar{A}$)	0	1	1	0	0

Activity 3 – Verilog stepper motor controller

Objective: write Verilog code to control the motion of a stepper motor.

1. Open Quartus II and start the New Project Wizard.
 - Working directory: fpga; project name: activity1; top-level entity: step.
 - Device family: Cyclone II (EP2C35F672C6N) or Cyclone IV (EP4CE115F29C7N).
 - Accept the default EDA tool settings and finish.
2. Create a new Verilog HDL file and enter the code in Listing 4.3 (step, divider, and stepper modules).
3. Save as step and start compilation.
4. Use **Tools** → **Netlist Viewer** → **RTL Viewer** to generate the block overview shown in Figure 4.3.

5. Assign pin numbers as shown in Table 4.5 (refer to the DE2 user manual). For the phaseout outputs that interface with the stepper motor driver, use expansion header GPIO_0.
6. Download the program to the Altera DE2 board.
7. Discuss the result.

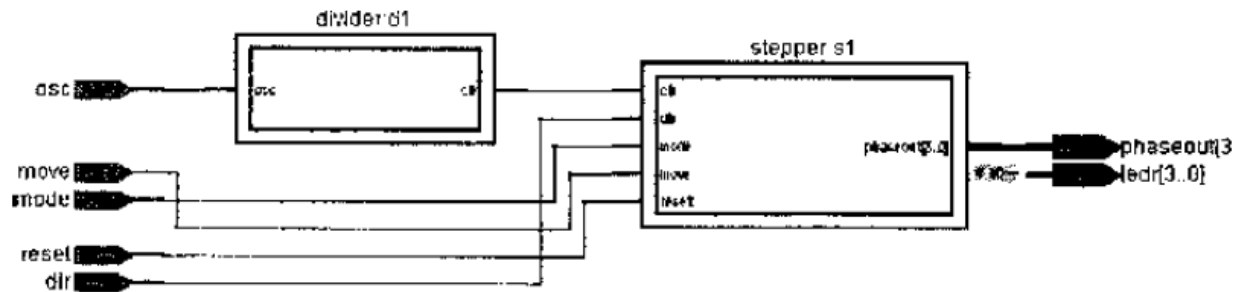


Figure 4.3: RTL Viewer overview of the step top-level module.

Table 4.5: Pin assignment for stepper motor control.

#	Node name	Direction	Pin location	#	Node name	Direction	Pin location
1	osc	Input		5	ledr[0]	Output	
	dir (SW0)	Input		6	phaseout[3]	Output	
	mode (SW1)	Input		7	phaseout[2]	Output	
	move (SW2)	Input		8	phaseout[1]	Output	
	reset (SW3)	Input		9	phaseout[0]	Output	
2	ledr[3]	Output					
3	ledr[2]	Output					
4	ledr[1]	Output					

step.v – top-level module

```

1  //=====
2  // Stepper Motor Controller
3  //=====
4  // TOP LEVEL MODULE
5  // reset (active low) brings the shaft to its reference position.
6  // mode selects step (1) or continuous (0) operation.
7  // dir selects clockwise (1) or anti-clockwise (0) motion.
8  // move shifts the shaft by one half-step in step mode.
9  module step(osc, reset, dir, mode, move, phaseout, ledr);
10     input  osc, reset, mode, dir, move;
11     output [3:0] phaseout, ledr;
12
13     wire clk;
14
15     divider d1 (
16         .osc (osc),
17         .clk (clk)
18     );
19
20     stepper s1 (
21         .clk      (clk),
22         .reset    (reset),
23         .dir      (dir),
24         .mode     (mode),
25         .move     (move),
26         .phaseout (phaseout)
27     );
28 endmodule

```

step.v – motor controller module (part 1 of 2)

```

1  // Module stepper is the motor controller.
2  module stepper(clk, reset, dir, mode, move, phaseout);
3     input  clk, reset, dir, mode, move;
4     output [3:0] phaseout;
5     reg [3:0] phaseout, position, ledr;
6
7     always @(posedge clk or negedge reset)
8     begin
9         if (reset == 0) begin
10             // motor resets to phase 1000 when reset is asserted
11             phaseout = 4'b1000;
12             ledr      = phaseout;
13         end
14         else begin
15             if (mode == 0) begin
16                 // Mode = 0: continuous motion.
17                 // The 8-state sequence below follows Table 1 in both
18                 // directions; only the first two states are shown here,
19                 // the remaining six states (0100, 0110, 0010, 0011,
20                 // 0001, 1001) follow the identical if(dir==0)/else

```

```

21 // pattern, cycling back to 1000.
22 case (phaseout)
23     4'b1000: if (dir == 0) begin
24         phaseout = 4'b1100; ledr = phaseout;
25     end else begin
26         phaseout = 4'b1001; ledr = phaseout;
27     end
28     4'b1100: if (dir == 0) begin
29         phaseout = 4'b0100; ledr = phaseout;
30     end else begin
31         phaseout = 4'b1000; ledr = phaseout;
32     end
33     // ... 4'b0100, 4'b0110, 4'b0010, 4'b0011,
34     //      4'b0001, 4'b1001 follow the same pattern ...
35     default: $display("invalid input");
36 endcase
37 end
38 else begin
39     // Mode = 1: single-step motion, advances only when
40     // move == 0. Uses the same 8-state sequence as above,
41     // indexed by `position` instead of `phaseout`.
42     if (move == 1'b0) begin
43         case (position)
44             // ... identical 8-state sequence as continuous
45             // mode, see Table 1 ...
46             default: $display("invalid input");
47         endcase
48     end
49 end
50 end
51 end

```

step.v — motor controller module (part 2 of 2)

```

1 // Motor advances by one half-step on the falling edge of move.
2 always @(negedge move or negedge reset)
3 begin
4     if (reset == 0) begin
5         position = 4'b1000; // NOTE: corrected from the original
6                             // listing, which printed this value
7                             // as "4.101011" -- likely a
8                             // transcription error; 4'b1000
9                             // matches the reset state used
10                            // consistently elsewhere.
11         ledr = position;
12     end
13     else if (reset == 1) begin
14         position = phaseout;
15         ledr = phaseout;
16     end
17 end
18 endmodule

```

step.v – clock divider module

```

1 // Module divider divides the incoming clock to produce a
2 // clock suitable for driving the motor controller.
3 module divider(osc, clk);
4     input  osc;
5     output clk;
6     reg    clk;
7     reg [15:0] count;
8
9     initial count = 16'b0000000000000000;
10
11    always @(posedge osc)
12    begin
13        count = count + 1;
14        clk    = count[15];
15    end
16 endmodule

```

Listing 4.3: step.v – top-level, motor-controller, and clock-divider modules.

Activity 4 – Variable-speed control with display feedback

Objective: extend the controller with selectable speed and direction/mode feedback.

1. Write Verilog code that allows selection of three different speeds for the stepper motor in continuous-motion mode.
2. Display the motor's direction of rotation on the seven-segment display, and the current mode (step / continuous) on the LCD display.
3. Print and submit the Verilog code and schematic diagram.

Appendix: Stepper Motor Datasheet Reference

Relevant specifications and wiring information for the Parallax #27964 stepper motor used in this experiment, extracted from the manufacturer datasheet (see References).



Figure 4.4: Parallax #27964 stepper motor.

Table 4.6: Technical specifications, Parallax #27964 stepper motor.

Parameter	Value
Type	4-phase, 12 V unipolar
Step angle	3.6° per step
Phase resistance	75 Ω
Current	150 mA
Phase inductance	39 mH
Detent torque	80 g·cm
Holding torque	600 g·cm
Mounting hole spacing (diagonal)	1.73 in
Mounting hole diameter	0.11 in
Shaft diameter	0.197 in
Shaft length	0.43 in
Motor diameter	1.66 in
Motor height	1.35 in
Weight	0.55 lb

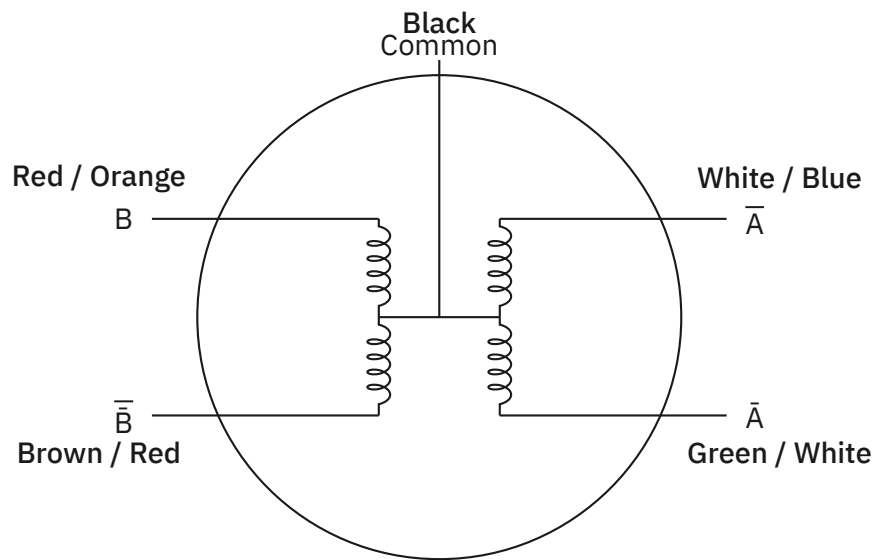


Figure 4.5: Coil winding and lead-wire colours, Parallax #27964 stepper motor.

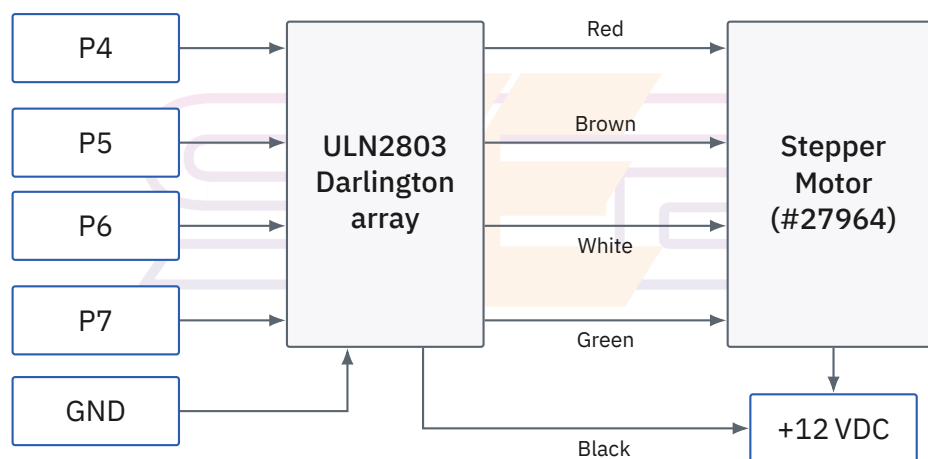


Figure 4.6: Reference driver-wiring diagram.

Note:

1. The P4–P7 labelling above follows the datasheet’s original microcontroller example; in this experiment, the corresponding driver inputs are instead the phaseout [3:0] outputs of the stepper module, assigned to DE2 GPIO pins as listed in Table 4.5.
2. The GND are to be common ground with the DE2 board, ULN2803 and the power source.
3. The +12 VDC is to be supplied from an external power source, not the DE2 board.

References

1. Altera DE2 User Manual.
2. *Introduction to Quartus II Software – Verilog, VHDL, Schematic*, Altera University Program.

3. *Introduction to Verilog*, Carleton University course notes.
4. *Stepper Motor Controller Using MAX II CPLDs*, Altera Application Note AN-488.
5. Parallax Inc., *4-Phase 12-Volt 75 Ω Unipolar Stepper Motor (#27964)*, technical datasheet.



UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering
Universiti Sains Malaysia, Engineering Campus

SEMESTER 1, 2026/2027

EEM 441

Control & Instrumentation Laboratory

Experiment 5

INTERFACING FPGA TO AN ANALOG-TO-DIGITAL
CONVERTER

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 5

Interfacing FPGA to Analog-to-Digital Converter

Objectives

1. To understand the operation of an analog-to-digital converter (ADC).
2. To interface an FPGA, via the Altera DE2 board, to an ADC.

Equipment / Software Required

1. Altera Quartus II software
 - version 9/10/12 for Cyclone II
 - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Multimeter / oscilloscope
4. ADC0804 analog-to-digital converter
5. Resistors: 10 k Ω
6. Variable resistor (potentiometer), 10 k Ω
7. Capacitor: 150 pF
8. 8 LEDs
9. 74LS245 octal bus transceiver

Introduction

This experiment explores interfacing an ADC chip to an FPGA. ADCs are among the most widely used data-acquisition devices: they translate an analog signal into a digital number that a processor (or, here, an FPGA) can read. A widely used ADC is the ADC0804 (National Semiconductor): it operates from +5 V and has 8-bit resolution.

Besides resolution, *conversion time* is a major factor in judging an ADC: the time the ADC takes to convert its analog input to a digital number. For the ADC0804 this depends on the clock signal applied to the CLK R / CLK IN pins, but cannot be faster than 110 μ s.

The ADC0804's input range is set by the $V_{ref}/2$ pin, as shown in Table 5.1. The chip's outputs are 5 V logic; since the DE2 board's inputs only tolerate 3.3 V, an octal bus transceiver (74LS245) must be used to step the output voltage down before it reaches the FPGA.

$$D_{out} = \frac{V_{in}}{\text{step size}}$$

where D_{out} is the digital data output and V_{in} is the analog input voltage.

Data conversion with the ADC0804 follows this sequence (Figure 5.1):

- Set $\overline{CS} = 0$ and send a low-to-high pulse on the $\overline{WR}$ pin to start the conversion.
- Monitor the $\overline{INTR}$ pin: while it is high, the conversion is still in progress; once it goes low, the conversion is complete.
- With $\overline{INTR}$ low, set $\overline{CS} = 0$ again and send a high-to-low pulse on the $\overline{RD}$ pin to read the data out of the ADC0804.

Table 5.1: $V_{ref}/2$ relation to V_{in} range.

$V_{ref}/2$ (V)	V_{in} range (V)	Step size (mV)
Not connected	0 to 5	$5/256 = 19.53$
2.0	0 to 4	$4/255 = 15.62$
1.5	0 to 3	$3/256 = 11.71$
1.28	0 to 2.56	$2.56/256 = 10.00$
1.0	0 to 2	$2/256 = 7.81$
0.5	0 to 1	$1/256 = 3.90$

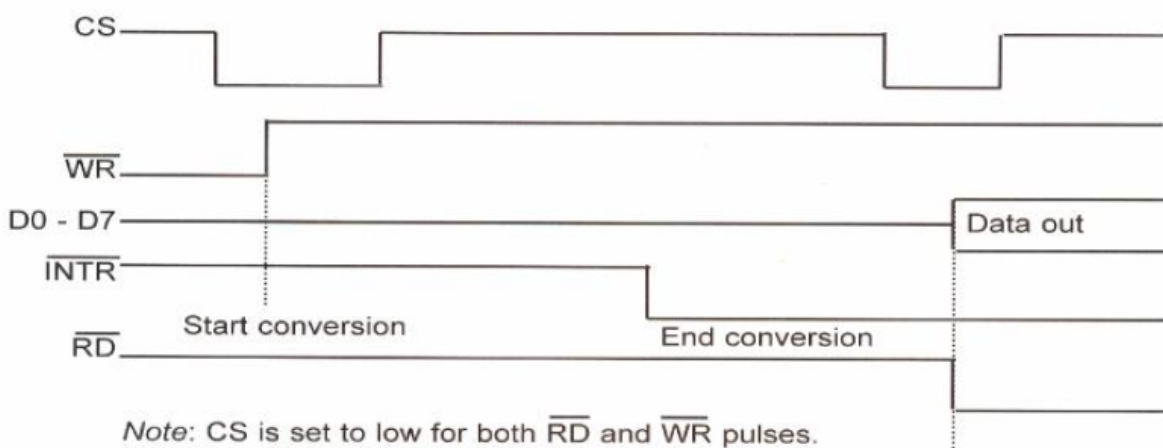


Figure 5.1: Read and write timing for the ADC0804.

Procedure

Activity 1 – Free-running mode

Objective: run the ADC0804 in free-running mode.

- Connect the ADC0804 as shown in Figure 5.2 for free-running mode.

2. Vary the input signal from 0 to 5 V using the 10 k Ω potentiometer shown in Figure 5.2 as the ADC's input. The input must not exceed 5 V.
3. Tabulate your results in Table 5.2.
4. Compare your hardware result with the value calculated using Equation (1).

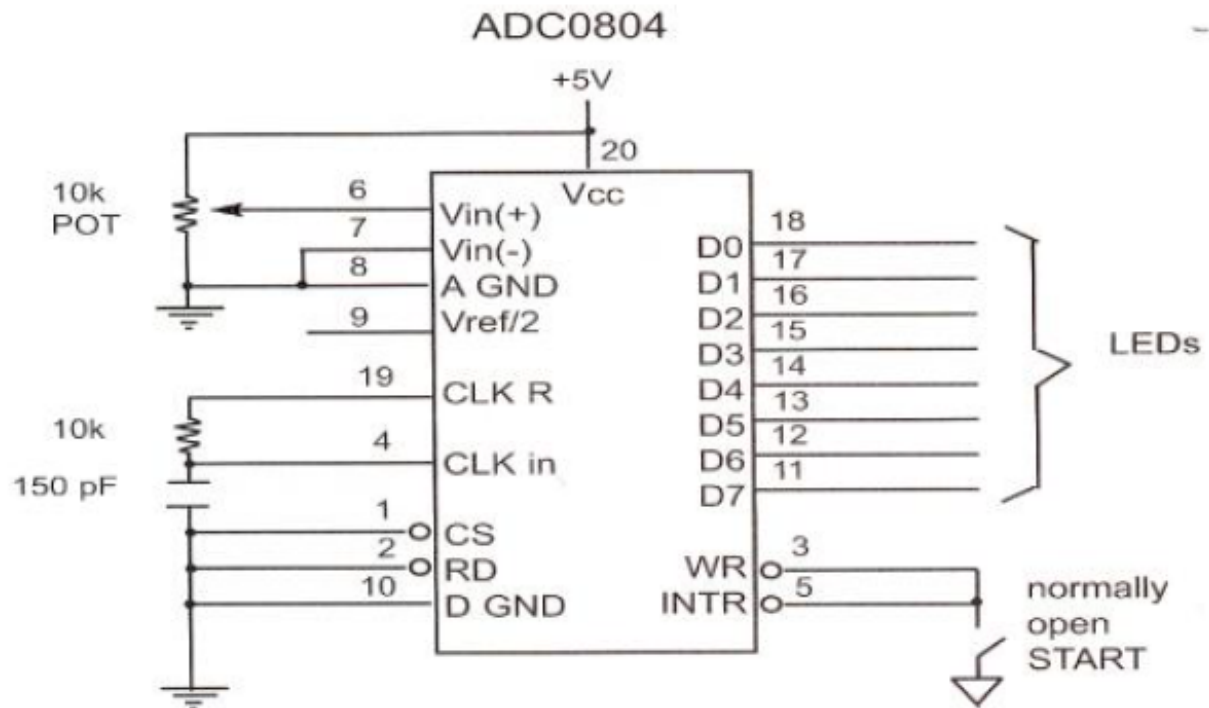


Figure 5.2: ADC0804 test circuit for free-running mode.

Table 5.2: Free-running mode output.

#	Input (V)	Output (hardware)	Output (calculation)
1	0.0		
2	0.5		
3	1.0		
4	1.5		
5	2.0		
6	2.5		
7	3.0		
8	3.5		
9	4.0		
10	4.5		

Activity 2 – Interfacing the ADC to the FPGA, displaying on LEDs

Objective: interface the ADC to the FPGA and display the digitised result on LEDs.

1. Use the circuit from Figure 5.2, modified as follows.
2. Connect the $\overline{RD}$ and $\overline{WR}$ pins to general-purpose I/O pins on the DE2 board, declared as **outputs**.
3. Connect the $\overline{INTR}$ and D0–D7 pins to the inputs of a 74LS245. Connect the 74LS245's outputs to general-purpose I/O pins on the DE2 board, declared as **inputs**. Refer to the DE2 user manual for pin locations.
4. In Verilog, generate the $\overline{WR}$ pulse to start conversion, poll $\overline{INTR}$ for end of conversion, then pulse $\overline{RD}$ to read the data on D0–D7.
5. Assign the FPGA's digitised outputs to the LED pins.
6. Compile, simulate, and download the program to the DE2 board.
7. Use the potentiometer to vary the input value.
8. Tabulate your results in Table 5.3.
9. Print and submit the Verilog code, simulation result, and experimental results.

Table 5.3: FPGA-interfaced ADC output.

#	Input (V)	Output
1	0.8	
2	1.7	
3	2.3	
4	2.8	
5	3.1	
6	3.6	
7	3.9	
8	4.1	
9	4.6	
10	4.8	

Activity 3 – Displaying the ADC result on the LCD

Objective: interface the ADC0804 to the DE2 board and display the result on the LCD.

1. Using the same circuit as in Activity 2, write Verilog code to read data from the ADC and display the result on the LCD (see Experiment 5 for the LCD interface).
2. Print and submit the Verilog code and simulation result.



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Control & Instrumentation Laboratory

Experiment 6

INTERFACING FPGA TO A DIGITAL-TO-ANALOG CONVERTER

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 6

Interfacing FPGA to Digital-to-Analog Converter

Objectives

1. To understand the operation of a digital-to-analog converter (DAC).
2. To interface an FPGA, via the Altera DE2 board, to a DAC.

Equipment / Software Required

1. Altera Quartus II software
 - version 9/10/12 for Cyclone II
 - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Multimeter / oscilloscope
4. DAC0800 digital-to-analog converter
5. Resistors: 10 k Ω ($\times 4$), 100 Ω ($\times 1$)
6. Capacitor: 0.01 μF

Introduction

A digital-to-analog converter (DAC) converts digital pulses into an analog signal. The first criterion for judging a DAC is its resolution, which depends on the number of binary inputs: the number of distinct analog output levels equals 2^n , where n is the number of data-bit inputs. An 8-bit DAC such as the DAC0800 therefore provides 256 discrete voltage (or current) output levels.

A DAC can be used to generate a sine wave. The output voltage for a given angle θ is:

$$V_{out} = 5\text{ V} + (5\text{ V} \times \sin \theta)$$

To find the digital value to send to the DAC for a given angle, multiply the resulting V_{out} by 25.6 (since there are 256 steps across a 10 V full-scale output).

Procedure

Activity 1 – DAC0800 operation

Objective: understand the operation of the DAC0800.

1. Connect the DAC0800 as shown in Figure 6.1.

2. Apply the digital input values at D0–D7 as listed in Table 6.1.
3. Measure the resulting analog output and tabulate your results in Table ??.

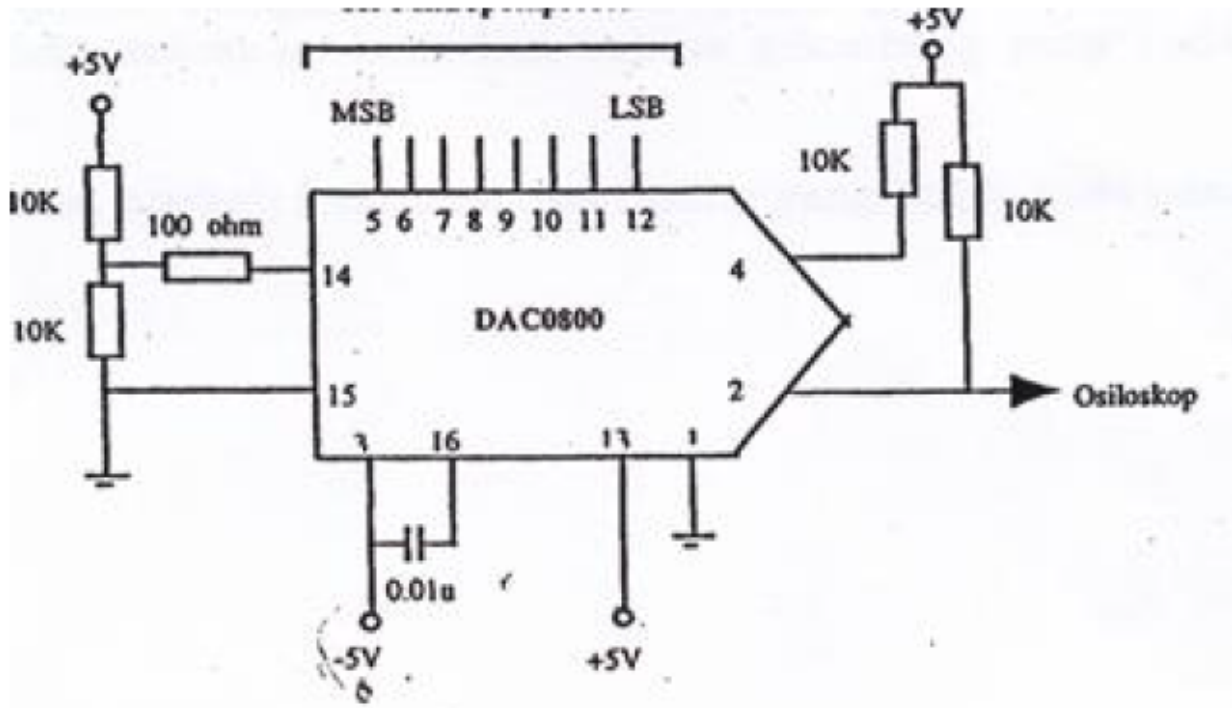


Figure 6.1: DAC0800 test circuit with op-amp output stage.

Table 6.1: DAC0800 digital-to-analog conversion results.

Digital value (decimal)	Analog value
0	
10	
60	
80	
100	
128	
180	
200	
220	
255	

Activity 2 – Generating a square wave

Objective: interface the FPGA to the DAC0800 and generate a square wave.

1. Connect the circuit as shown in Figure ??.
2. Connect D0–D7 of the DAC to general-purpose I/O pins on the DE2 board.
3. Write Verilog code to generate a square wave with a frequency of 100 Hz and a 70% duty cycle.
4. You may use a lookup table, following the same approach as the LCD program in Experiment 5.
5. Print and submit the Verilog code and simulation result.

Activity 3 – Generating a sine wave

Objective: interface the FPGA to the DAC0800 and generate a sine wave.

1. Using the equation in the Introduction, complete Table 6.2.
2. Connect the circuit as shown in Figure ??.
3. Connect D0–D7 of the DAC to general-purpose I/O pins on the DE2 board.
4. Write Verilog code to generate a sine wave with a frequency of 100 Hz.
5. You may use a lookup table, following the same approach as the LCD program in Experiment 5.
6. Print and submit the Verilog code and simulation result.

Table 6.2: Angle vs. voltage magnitude for the sine wave.

Angle (°)	$\sin \theta$	V_{out} (V)	Value sent to DAC ($V_{out} \times 25.6$)
0			
30			
60			
90			
120			
150			
180			
210			
240			
270			
300			
330			
360			



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Experiment 7

DC MOTOR SPEED MEASUREMENT

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 7

Computer-Based Motor Speed Measurement

Objectives

1. To understand the principle and operation of computer-based measurement systems.
2. To learn to acquire and process signals automatically using a data-acquisition (DAQ) system.
3. To become familiar with signal conditioning for interfacing sensors with digital acquisition hardware.
4. To measure and analyse the rotational speed of a DC motor using pulse signals.
5. To develop skills in integrating hardware and software for real-time measurement and analysis.

Equipment / Software Required

- Personal computer (PC) running Windows, with a C/C++ development environment
- Data Acquisition (DAQ) card: Advantech PCI-1750 (16 isolated digital inputs, 16 isolated digital outputs, 1 isolated counter, 1 timer, 2500 VDC optical isolation), installed in the host PC, with driver and **Advantech Navigator.exe** installed
- Sensor trainer with DC motor and a 12-slot wheel-encoder pulse output
- Components for the student-designed filter and signal-conditioning circuit (op-amps, resistors, capacitors, Schmitt-trigger ICs, optocouplers)
- Digital storage oscilloscope (DSO), dual-channel, ≥ 50 MHz, with passive probes
- Connecting wires and breadboard

Introduction

In most measurement processes it is often necessary to acquire data automatically, without a human operator. A computer-based instrument has several advantages over conventional measuring instruments: it can acquire measurements quickly and accurately, sustain real-time control, and operate for extended periods with consistent precision.

A computer-based measurement system typically contains the following elements (Figure 7.1): a transducer, a signal-conditioning stage, data-acquisition hardware, and software running on the host PC.

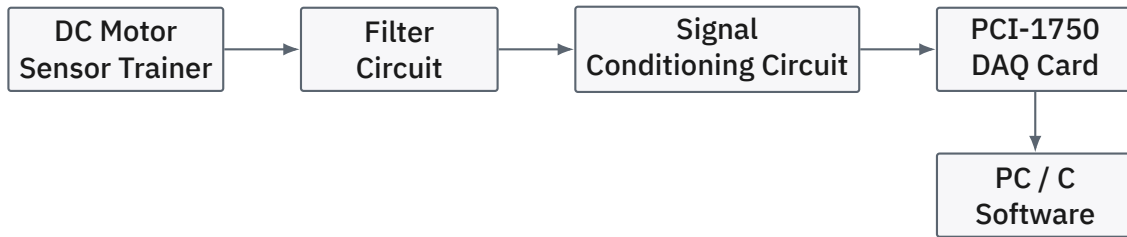


Figure 7.1: Computer-based measurement system: transducer, signal conditioning, DAQ hardware, and analysis software.

Safety & Hardware Protection

1. **Maximum voltage tolerance (0–5 V TTL limit).** The PCI-1750's digital input channels require standard 0–5 V TTL logic levels. Never connect a raw sensor signal exceeding 5 V directly to the DAQ card input. Measure the raw pulse voltage amplitude with the DSO *before* making any DAQ card connection.

Procedure

Activity 1 – Characterising the encoder signal and designing the filter

Objective: measure the encoder's pulse characteristics and design a signal-conditioning filter.

1. Power on the sensor trainer and verify motor rotation. Wire the encoder's raw output test point to Channel 1 of the oscilloscope.
2. Run the motor across its usable speed range and observe the encoder pulse train on the oscilloscope. Record the peak-to-peak voltage, rise time, overshoot, and any noise ripple on the raw signal.
3. At a representative operating speed, measure the pulse period and estimate the corresponding fundamental frequency from the oscilloscope trace.
4. From this measurement, determine a suitable cutoff frequency for the noise-rejection filter.
5. Design the necessary filter based on the cutoff frequency determined above. Justify your design choices.
6. Design the signal-conditioning circuit needed to present the filtered encoder pulses to the PCI-1750's digital input with valid 0–5 V TTL logic levels. Justify your design choices.

Activity 2 – Interfacing to the DAQ hardware

Objective: connect the conditioned encoder signal to the PC via the PCI-1750, and verify the connection before writing any software.

1. With the motor trainer powered off, connect the output of your signal-conditioning circuit to the designated digital input channel (e.g. DIO / Counter In) on the PCI-1750 terminal board. Connect a common ground between your breadboard and the PCI-1750 terminal board ground.
2. On Channel 2 of the oscilloscope, verify that the conditioned signal is a clean 0–5 V square wave with sharp edges and no false-trigger spikes, while keeping Channel 1 on the raw input for comparison. Adjust your filter's RC components if ringing occurs.
3. Launch **Advantech Navigator.exe** on the PC to verify hardware channel detection and pulse-counting activity — use this strictly as a hardware diagnostic to confirm the expected pulses are being registered, before writing any code.
4. Close Navigator.exe and launch your C development environment.



Figure 7.2: Signal path from the DC motor trainer to the host PC.

Activity 3 – Speed measurement software

Objective: implement software to compute and display motor speed (frequency and RPM) in real time, using two selectable measurement methods.

Two complementary methods can be used to convert the pulse train into a frequency and rotational speed, and your software should let the user select between them:

Method 1 – Fixed-Time (by counts, $1 \leq N \leq 99$): the user specifies a target pulse count N ; the software measures the elapsed time T_N for N pulses to occur, then computes

$$f = \frac{N}{T_N \times P} \text{ [rev/s]}, \quad \text{RPM} = 60f, \quad \omega = 2\pi f \text{ [rad/s]}$$

where P is the encoder's pulse resolution (pulses per revolution).

Method 2 – Fixed-Count (by period, $1 \leq T \leq 9 \text{ s}$): the user specifies a measurement window T seconds; the software counts the number of pulses N_T that occur within that window, then computes

$$f = \frac{N_T}{T \times P} \text{ [rev/s]}, \quad \text{RPM} = 60f, \quad \omega = 2\pi f \text{ [rad/s]}$$

In both methods, t_{start} and t_{stop} (marking the beginning and end of the measurement) are obtained from the PCI-1750's counter/timer using

$$\text{Elapsed time} = \frac{t_{\text{stop}} - t_{\text{start}}}{\text{CLK_TCK}}$$

where CLK_TCK is the counter's clock tick rate.

1. In C, write a program that:
 - initialises the PCI-1750 via the vendor driver/SDK (counter and digital-input registers).
 - lets the user select the measurement method (Fixed-Time, $1 \leq N \leq 99$; or Fixed-Count, $1 \leq T \leq 9$ s) and enter the corresponding parameter.
 - for the *Fixed-Time* method: times how long it takes for N encoder pulses to occur.
 - for the *Fixed-Count* method: counts how many encoder pulses occur within the window T .
 - computes the frequency, RPM, and angular speed ω using the relations above, updating at least every 500 ms.
 - displays the running value (frequency and RPM, with the active method indicated) in a simple HMI, updating in real time, with out-of-range error detection.
2. Interface your filter/conditioning circuit and the PCI-1750 together and perform measurements across the motor's input voltage range (e.g. 3.0 V to 12.0 V in 1.5 V increments), using both methods at each operating point. Compare your computed speed against a direct oscilloscope measurement of the encoder period at the same operating point.
3. Tabulate your results (Tables 7.2–7.3).
4. Plot computed speed (from both software methods) against the oscilloscope-measured speed, and plot relative error (%) against motor speed for both methods.
5. Discuss and comment on your results, including any discrepancy between the two software methods and the oscilloscope reference, and how each method's resolution changes with motor speed (see the trade-off noted above).

Table 7.1: By-counts method: frequency comparison table.

N	Measured Frequency (Hz)					Oscilloscope (Hz)
	#1	#2	#3	#4	#5	

Table 7.2: By-counts method — RPM comparison table.

N	ω (Measured RPM)					Oscilloscope (calculated RPM)
	#1	#2	#3	#4	#5	

Table 7.3: By-period method: frequency comparison table.

T (s)	Measured Frequency (Hz)					Oscilloscope (Hz)
	#1	#2	#3	#4	#5	

Table 7.4: By-period method: RPM comparison table.

T (s)	ω (Measured RPM)					Oscilloscope (calculated RPM)
	#1	#2	#3	#4	#5	

Report

1. Present your oscilloscope measurements of the raw and conditioned encoder pulse, and your resulting filter design (topology, calculations, and component values), together with the measured response of the built filter.
2. Present your design of the signal-conditioning stage between the filter and the PCI-1750 digital input, with justification, including the isolation and grounding measures you applied (see Safety & Hardware Protection above).
3. Compare your software-computed speed (both Fixed-Time and Fixed-Count methods) with the oscilloscope-measured speed across your tested operating points, and discuss the sources of any discrepancy.
4. Compare the Fixed-Time and Fixed-Count methods directly: at what motor speeds does each method give better resolution or a more stable reading, and why? Which would you recommend for a low-speed application, and which for a high-speed one?

5. What is the theoretical maximum measurable speed of this system, given the encoder resolution and your software's timing resolution?
6. Identify the primary sources of measurement error in your system (e.g. pulse jitter, counter quantisation error, sampling clock drift, motor speed ripple), and discuss their impact on overall RPM accuracy.

Deliverables

1. **C source code file (.c):** fully commented C software implementing pulse counting, timing calculations, RPM conversion, and HMI display.
2. **Technical design report (.pdf):** formal lab report detailing circuit schematics, component calculations, signal waveform captures, data tables, and error analysis.
3. **System demonstration:** a live bench demonstration showing real-time motor speed tracking on your C software HMI, alongside DSO verification.

References

- Advantech PCI-1750/PCI-1750SO user manual.



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EEM 441

Control & Instrumentation Laboratory

Experiment 8

TEMPERATURE MEASUREMENT AND CONTROL

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 8

Temperature Control Using Raspberry Pi

Objectives

1. To interface a temperature sensor and a heating element with a Raspberry Pi.
2. To implement a temperature control algorithm to maintain a target temperature.
3. To log temperature data and analyse system performance.

Equipment / Software Required

- Raspberry Pi 4B
- NTC thermistor (or other temperature sensor) and trainer board with indicator LED
- Analog-to-digital converter, **ADC0804**
- Digital-to-analog converter, **DAC0800**
- Operational amplifier, LM741
- Relay
- Resistors, capacitors (according to your design)
- NPN transistor, 2N3904
- LEDs
- Power supply and connecting wires

Introduction

In most industrial processes, there are possibilities when the temperature needs to be regulated automatically. For example, the temperature of fluid inside a reactor has to be controlled within a certain range in order to get a stable operation. In logic control, the temperature is controlled by switching the heater on or off when a certain critical value is reached, whereas fuzzy control needs a true definition of “hot” before the control operation.

In this open-ended lab, students will design and implement a system to perform temperature control using a Raspberry Pi. The goal is to create a complete system that captures the temperature sensor reading, processes the information, displays the temperature, and turning a LED on or off based on the temperature.

An overall block diagram for the measurement system is shown in Figure 8.1.

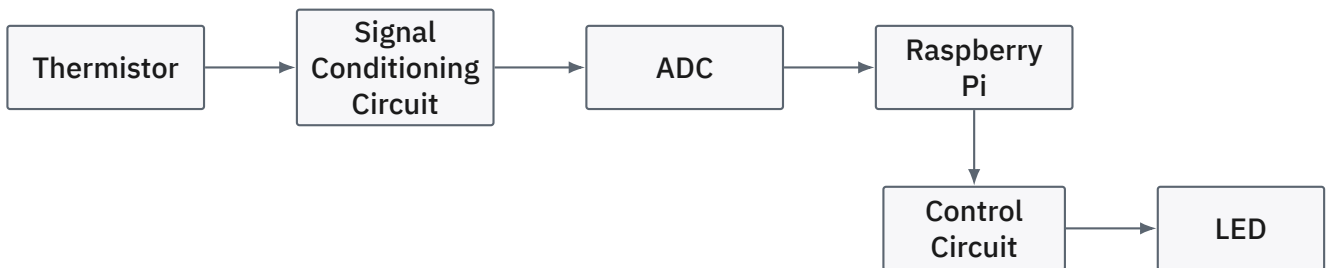


Figure 8.1: Block diagram of the temperature control system.

Procedure

Activity 1 — Signal-conditioning circuit design

Objective: design and build a circuit that converts the thermistor's resistance to a voltage suitable for the ADC0804 input.

1. Obtain the resistance–temperature characteristic for the NTC thermistor from its datasheet, and determine the resistance range corresponding to the temperature range you intend to measure and control.
2. Design a signal-conditioning circuit that converts the thermistor's resistance to a voltage. Calculate your component values and justify your design choices, including the expected output voltage range over your chosen temperature range.
3. Include a voltage-follower (buffer) stage, based on the LM741, to eliminate loading effects between the conditioning circuit and the ADC0804 input.
4. Build the circuit and verify its output voltage against a reference temperature measurement at two or more points, and check linearity/consistency with your design calculations.

Activity 2 — Configure and test the ADC and DAC

Objective: build and verify the ADC0804 and DAC0800 circuits in isolation, before integrating them with the Raspberry Pi.

1. Configure and construct the ADC0804 in free-running mode, with its input connected to a potentiometer and its outputs connected to LEDs. Test that the LED pattern changes correctly as the potentiometer is adjusted.
2. Configure and construct the DAC0800 circuit with switches connected to its digital inputs. Test the analog output with a multimeter as you toggle the switches, confirming the two ON/OFF output levels you intend to use.

Activity 3 — Interfacing to the Raspberry Pi

Objective: connect the ADC0804 and DAC0800 to the Raspberry Pi via a bidirectional level shifter, and connect the DAC0800's output to the heater control circuit.

1. Connect the output of your signal-conditioning circuit (Activity 1) to the V_{in} input of the ADC0804.
2. Wire the ADC0804's 5 V-side signals ($D0-D7$, $\overline{INTR}$, $\overline{CS}$, $\overline{RD}$, $\overline{WR}$) to the high-voltage (HV) side of a bidirectional level-shifter module, and the corresponding Raspberry Pi GPIO pins to the low-voltage (LV) side. Repeat for the DAC0800's digital inputs.
3. Connect the DAC0800's analog output to your heater control circuit (relay driven via the NPN 2N3904 transistor), which in turn drives the trainer's heater/indicator LED.
4. Verify both the input and output wiring with a simple test script before writing the full control program: confirm a static ADC reading is read back correctly, and confirm the heater/LED can be toggled on and off by writing to the DAC0800.

Activity 4 — Temperature monitoring and heater control software

Objective: implement software that displays the measured temperature and controls the heater according to a threshold.

1. In Python or C/C++, write a program that:
 - periodically triggers an ADC0804 conversion and reads back the digitised thermistor voltage,
 - converts the reading to a temperature using your circuit's calibration from Activity 1,
 - displays the current temperature in a simple terminal or GUI display, updating in real time, and
 - compares the measured temperature against the threshold and writes the corresponding ON/OFF level to the DAC0800 to control the heater.
2. Report and justify your design choice for the signal-conditioning circuit.
3. Use the flowchart in Figure ?? as a starting point for your control logic, adapting the ON/OFF branch to match your chosen design.
- 4.
5. Set the controlled (threshold) temperature to, for example, 20 °C, and record the observed thermostat temperature for 35 minutes using a 1-minute time interval (Table 8.1). Plot temperature against time.
6. Repeat for 23 °C and 25 °C. Plot your results.
7. Discuss your findings, including the responsiveness of the heater control and the accuracy of your temperature conversion.

Table 8.1: Temperature and heater-state log.

Threshold temp.	Time (min)	Measured temp.	Heater state
	1		
	2		
	...		
	35		

Report

1. Present your signal-conditioning circuit design (topology, calculations, component values) and its calibration against a reference temperature measurement.
2. Describe your ADC0804 and DAC0800 interfacing to the Raspberry Pi, including your level-shifting solution and why a unidirectional buffer would not be sufficient here.
3. State and justify your chosen heater ON/OFF logic.
4. Present and discuss your temperature-vs-time plots for each threshold tested (20 °C, 23 °C, 25 °C).
5. Assess the system's accuracy (e.g. ± 1 °C margin), response time after a setpoint change, and stability (absence of oscillation/overshoot).
6. What sources of error or lag affect the accuracy and responsiveness of this system (thermistor thermal mass, signal-conditioning bandwidth, ADC/DAC conversion time, software polling rate, etc.)?

Deliverables

1. **Code:** a programming script that accurately measures and displays the temperature based on sensor inputs, and controls the heater accordingly.
2. **System setup diagram:** a diagram showing the placement and wiring of sensors to the Raspberry Pi.
3. **Documentation:** a report that explains the setup, the temperature control process, sample readings, discussion, and any challenges encountered.
4. **Demo:** a short presentation or video demonstrating the working system and the collected data.

References

- ADC0804 datasheet.
- DAC0800 datasheet.
- NTC thermistor datasheet: <http://www.farnell.com/datasheets/307623.pdf>



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SEMESTER 1, 2026/2027

EEM 441

Control & Instrumentation Laboratory

Experiment 9

PROJECT CREATION IN SCADA (IOT)

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

: _____ Matrix # _____

Experiment 9

IIoT Monitoring & Alarm Visualization for a PLC-Controlled Machine

Objectives

1. To understand the role of each layer in an IIoT monitoring pipeline: PLC, SCADA, database, and dashboard.
2. To configure communication between a Siemens S7-1200 PLC and Advantech WebAccess/SCADA.
3. To bridge SCADA tag data to a PostgreSQL database via ODBC, so that history is available outside WebAccess.
4. To design a web-based IIoT dashboard displaying real-time sensor status and visual alarms, using the WISE-IoTSuite/Dashboard application.
5. To demonstrate a complete monitoring pipeline, from a physical sensor to a live dashboard panel.

Equipment / Software Required

- MTA SkillLab i-Factory kit: conveyor sorting machine with Siemens SIMATIC S7-1200 CPU 1212C AC/DC/Rly, fitted with vibration, laser-distance, temperature/humidity, and part-counting sensors
- PC/laptop with Advantech WebAccess/SCADA V9.x installed (Project Node and SCADA Node)
- PostgreSQL, with the psqlodbc (x86) ODBC driver and pgAdmin4 installed
- ODBC Data Source Administrator (32-bit) — WebAccess only supports 32-bit ODBC connections
- Ethernet cable and a configured local network connection between the PC and the PLC

Introduction

Industrial IoT (IIoT) monitoring bridges field-level operational technology (OT) – sensors, actuators, and the PLC that controls them – with IT-side visualization and analytics. This experiment builds that bridge one layer at a time, using the same i-Factory conveyor rig and WebAccess/SCADA software introduced in Experiment 5.

Communication protocols used at each layer (Table 9.1): Modbus links the sensors to the PLC; TCP/IP (or OPC UA) links the PLC to WebAccess/SCADA; and ODBC links WebAccess/SCADA to the PostgreSQL database that the Dashboard ultimately reads from.

The full pipeline is shown in Figure 9.1. Each stage depends on the one before it: a tag with no PLC communication shows no data; a tag with communication but no **Log Data** setting never reaches the database; and a database with no rows has nothing for the Dashboard to display. Troubleshooting this experiment usually means working backward through this chain to find where data stops flowing.

Table 9.1: Communication protocols in the IIoT pipeline.

#	Protocol	Role
1	Modbus	Communication between sensor and PLC
2	TCP/IP	Communication between PLC and WebAccess (SCADA)
3	ODBC	Communication between SCADA and the database
4	OPC UA	Alternative standard communication between PLC/equipment and SCADA



Figure 9.1: IIoT monitoring pipeline for this experiment.

Sensor Assignment

Each lab group is assigned one primary sensor/parameter to monitor for this experiment. Working with a distinct analog or discrete signal per group ensures each team independently validates its own signal path (PLC addressing, tag configuration, alarm thresholds, and dashboard binding) rather than inheriting settings tuned for a different sensor type.

Refer to Table 9.2 for your group's assigned sensor, its PLC DB address, and the suggested tag prefix to use in WebAccess/SCADA.

Table 9.2: Sensor assignment by group.

Group	Sensor / Parameter	PLC DB address	Suggested tag prefix
G1	Temperature	DB3, DBD92	G1_Temp
G2	Relative humidity	DB3, DBD88	G2_Humid
G3	Vibration — Z-axis RMS	DB3, DBD76	G3_VibZ
G4	Vibration — X-axis RMS	DB3, DBD72	G4_VibX
G5	Vibration sensor temperature	DB3, DBD80	G5_VibT
G6	Laser distance (analog)	DB3, DBD84	G6_Dist
G7	Good part count	DB3, DBD12	G7_GoodP
G8	Defect part count	DB3, DBD16	G8_DefP
G9	Machine OEE availability index	DB3, DBD24	G9_Avail
G10	Conveyor motor speed	DB3, DBD32	G10_MtrSpd

Every group also shares two common digital signals, regardless of assigned sensor: **Start** (DB1, DBX0.1), **Stop** (DB1, DBX0.3), and **Emergency Stop** (DB1, DBX0.0).

Include at least one of these as a discrete tag alongside your assigned analog sensor, so your dashboard has both a trend and a status indicator to display.

Procedure

Activity 1 — Create the SCADA project and connect to the PLC

Objective: create a WebAccess/SCADA project and bring your assigned sensor in as a live tag.

1. Open a web browser and navigate to 127.0.0.1 (or localhost) to reach the WebAccess/SCADA landing page. Select **Project Management** from the **Platform** dropdown and log in (default account admin, no password).

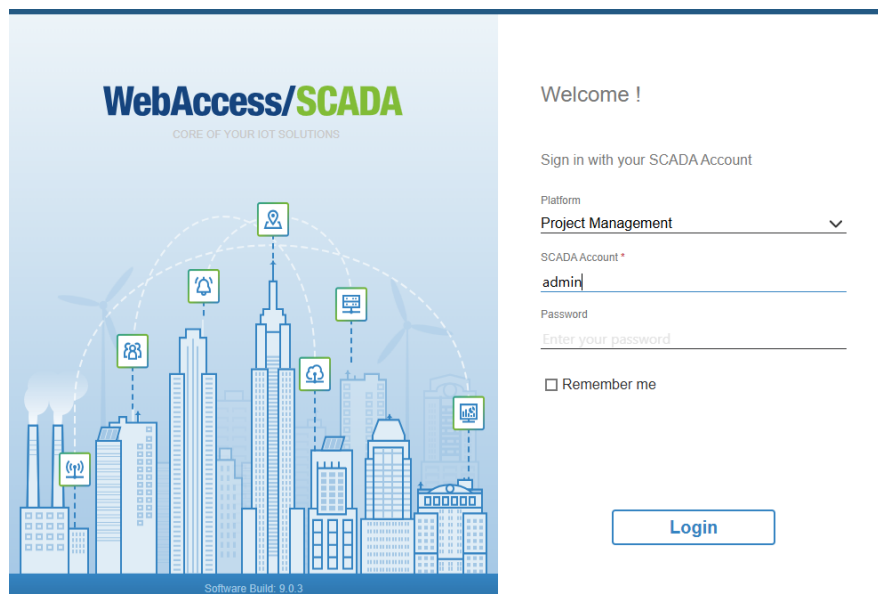


Figure 9.2: WebAccess/SCADA sign-in, with Project Management selected.

2. Create a new project: enter a **Project Name** and **Description**, set the **Project Node IP Address** (127.0.0.1 for local), and keep the default **Primary TCP Port** (4592) and **HTTP Port** (80) unless instructed otherwise.

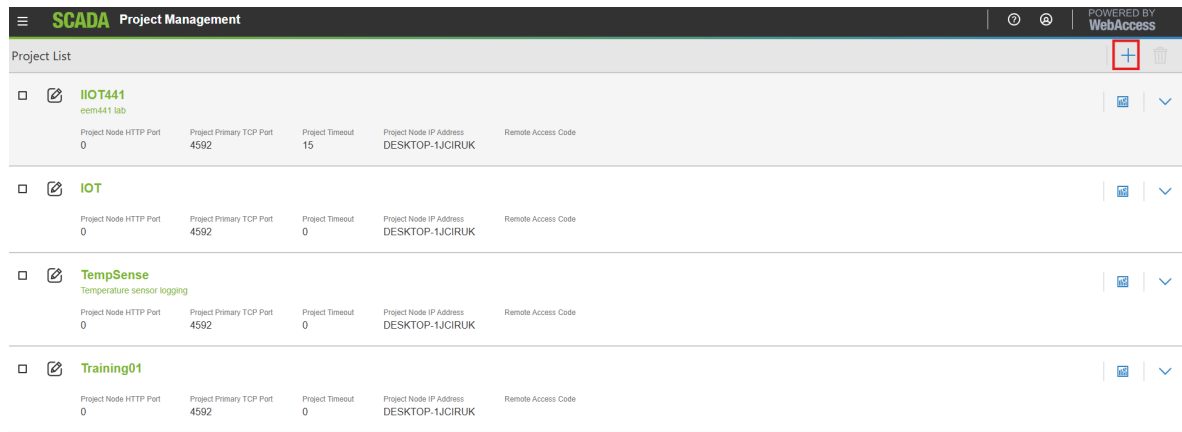


Figure 9.3: Create New Project form.

3. Create a **SCADA Node**: select your Project Node, click **+**, choose **New Node**, and fill in the Primary Node Name, TCP Ports, Node Timeout, and SCADA Node IP Address.



Figure 9.4: Create New SCADA Node form.

4. Add a **Communication Port**: choose **TCP/IP** as the interface, and configure the Scan Time (e.g. 1000 ms) and Auto-Recover Time.

Create New Comport
* Required

Interface Name	TCPIP
Comport Number *	1
Description	S7-1200 OEEE Monitoring TCP/IP
Scan Time *	1000
Timeout (Unit:MilliSecond) *	3000
Retry Count *	3
Auto Recover Time (Unit:Seconds) *	5

CANCEL SUBMIT

Figure 9.5: Communication port (Comport) settings.

5. Add a **Device** on that Comport: set Device Type to SiemS7, enter the PLC's IP address and port (102), and set the TSAP in hex for an S7-1200 (01.01).

Create New Device
* Required

Device Name *
Siemens PLC

Description
SIEMENS S7-1200

Unit Number *
0

Device Type
SiemS7

Manually switch Primary/Backup. *
0

TSAP in Hex :Device ID, RackSlot *
03.02

☐ Primary

CANCEL SUBMIT

Figure 9.6: Add Device form (S7-1200).

6. Create an IO Tag for your assigned sensor, using the prefix from Table 9.2 (e.g. G1_Temp), and a second discrete tag for Start, Stop, or Emergency Stop. Set the Address, Data Type, and Conversion Code to match your PLC DB offset.

Create New Tag
* Required

Basic Information

Tag Type
DBD (analog)

Tag Name *
TemperatureSensor

Description
DB DWord Data

Scan Type
Constant Scan

Address *
DBD3,96

Conversion Code
Unsigned Integer

Start Bit *
0

Length *
32

Signal Reverse
☐ Yes ☒ No

Scaling factor 1 *
0

Scaling factor 2 *
0

Scaling Type
No Scale

CANCEL SUBMIT

Figure 9.7: Insert IO Tag form (analog sensor example).

7. Start the SCADA Node's Kernel, open **ViewDAQ**, and confirm both tags are updating with live values that reflect what is actually happening on the machine.

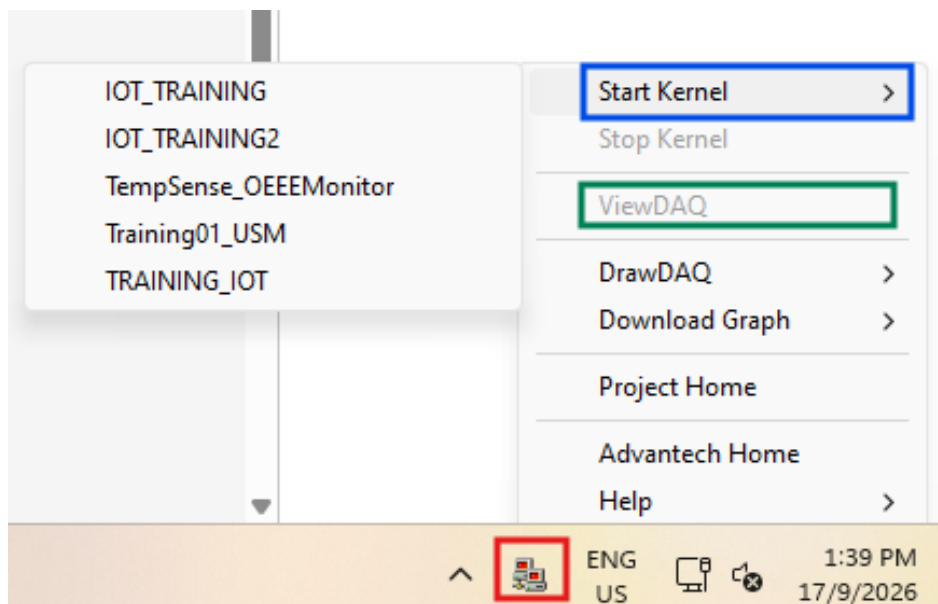


Figure 9.8: Kernel running, viewed in ViewDAQ.

Activity 2 – Configure an alarm

Objective: enable alarming on your assigned sensor, so threshold breaches are visible in both SCADA and, later, the dashboard.

1. Open your analog sensor tag's configuration and enable a non-zero alarm priority (High, High-High, Low, or Low-Low, as appropriate to your sensor).

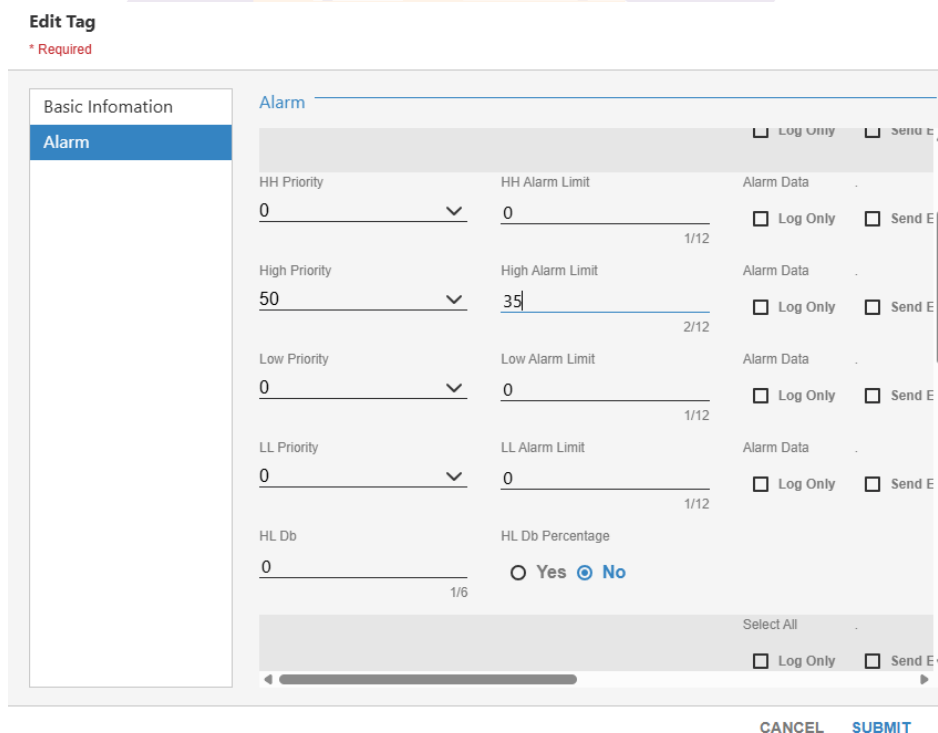


Figure 9.9: Alarm priority configuration.

2. Enable state alarming on your discrete tag (e.g. alarm when Emergency Stop becomes active).

3. In **ViewDAQ**, trigger a manual threshold breach (e.g. briefly block the sensor's path, or press the Emergency Stop) and confirm the alarm appears in the SCADA **Alarm Summary**.

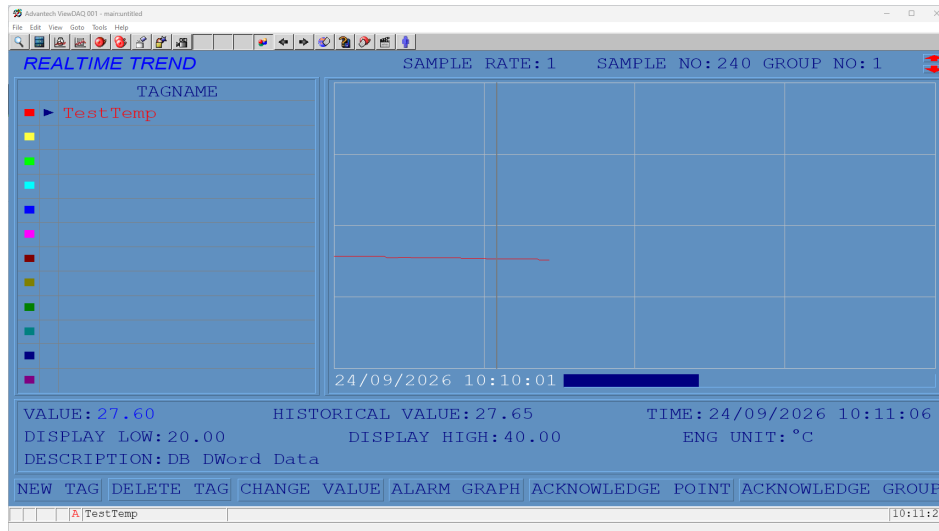


Figure 9.10: Alarm triggered, viewed on the real-time trend.

Activity 3 – Bridge SCADA to PostgreSQL via ODBC

Objective: connect WebAccess to a PostgreSQL database, and enable logging so your tag's history is written there.

1. Confirm with your instructor that PostgreSQL, the `psqlodbc` (x86) driver, and `pgAdmin4` are already installed — this experiment assumes they exist; installing them from scratch is outside the scope of this lab.
2. In `pgAdmin`, create a connection to the PostgreSQL server (Host `127.0.0.1`, Port `5436`, default credentials provided by your instructor), then create a new database for this experiment.

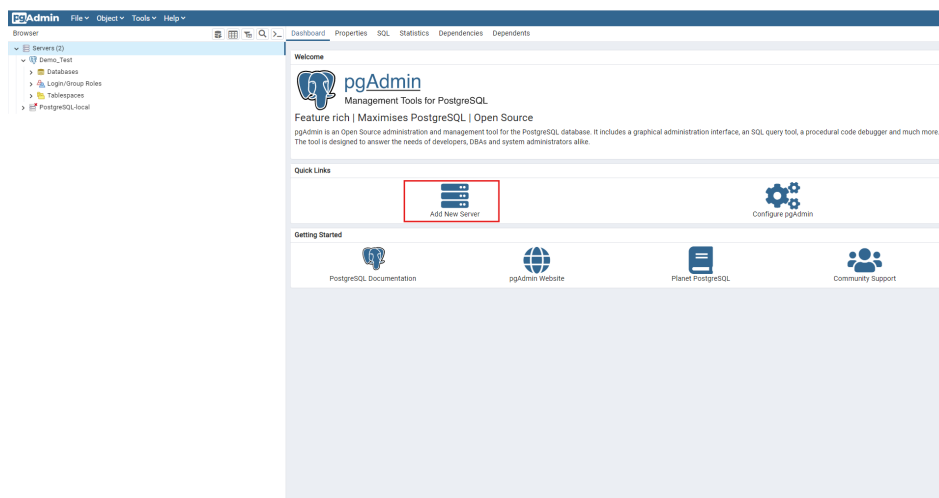


Figure 9.11: pgAdmin: registering the PostgreSQL server.

3. Open **ODBC Data Source Administrator (32-bit)**, switch to the **System DSN** tab, click **Add**, and select **PostgreSQL Unicode**. Fill in the Data Source name, Database,

Server, Port, Username, and Password to match the database you just created.

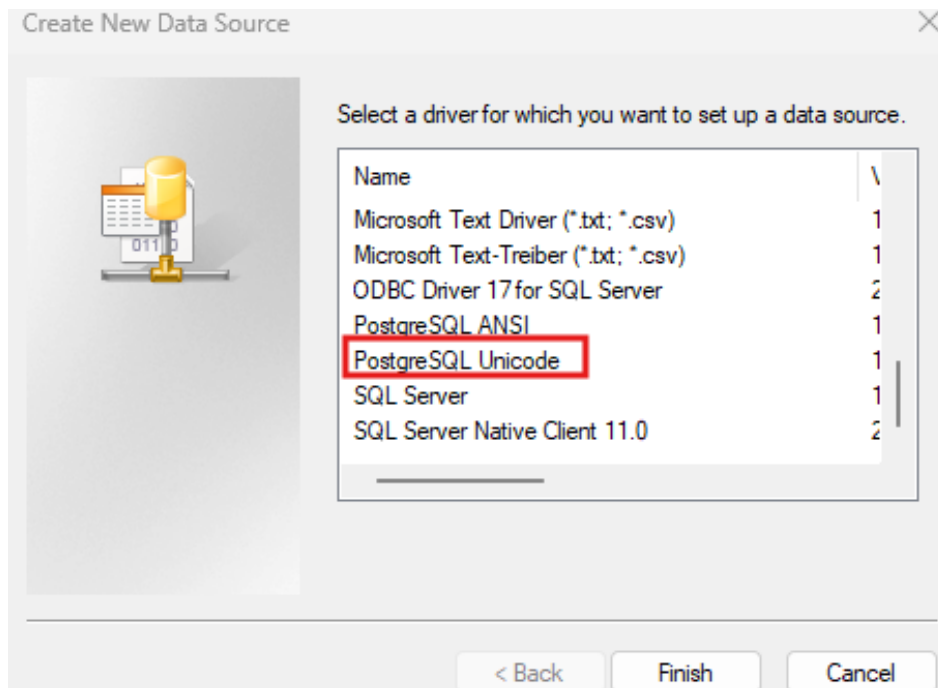


Figure 9.12: ODBC Data Source Administrator (32-bit).

- Back in WebAccess, open the project's **Setting** menu → **ODBC data record source**, and insert a new ODBC entry: a numeric Log Data Source ID, the DSN name from the previous step, and matching login credentials. Test the connection.

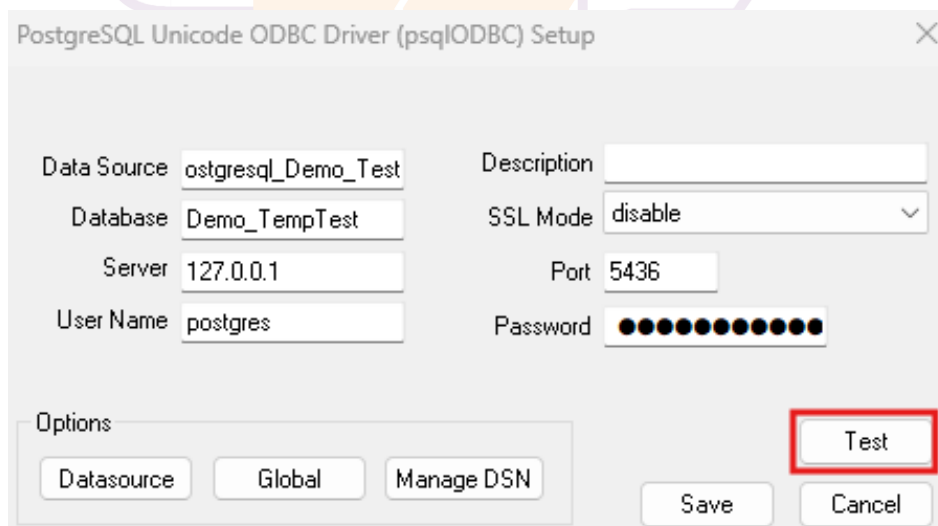


Figure 9.13: WebAccess ODBC data record source.

- On your SCADA Node, enable **Data Log to ODBC** (**Edit Node** → **Data Log**). Then, on each tag you want logged, enable **Log Data** and set the **Data Log Interval** to 0 (record by deadband) to avoid missed samples.

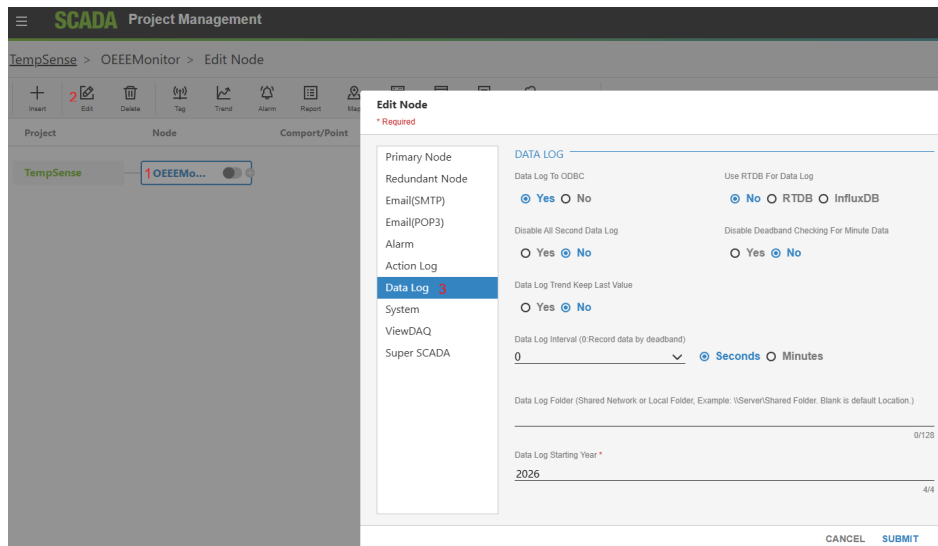


Figure 9.14: SCADA Node Edit Node: Data Log tab.

- Download the project to the SCADA Node, then confirm in pgAdmin that rows are appearing in the corresponding log table (e.g. bwanaLogtable for analog tags).

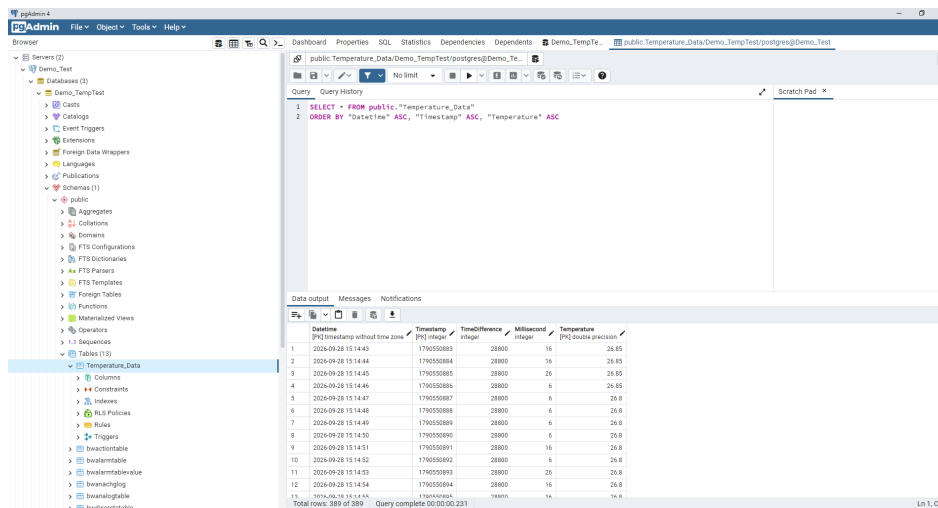


Figure 9.15: pgAdmin: logged rows in the analog log table.

Activity 4 – Build a dashboard

Objective: visualize your assigned sensor and its alarm state on a WISE-IoTSuite/Dashboard panel.

1. From the WebAccess landing page, select **Dashboard** from the **Platform** drop-down instead of Project Management.
2. In the Dashboard app, open **Configuration** → **Data Sources** → **Add data source**, choose **PostgreSQL**, and enter the same connection details used for the ODBC bridge in Activity 3. Click **Save & Test** and confirm a successful connection.

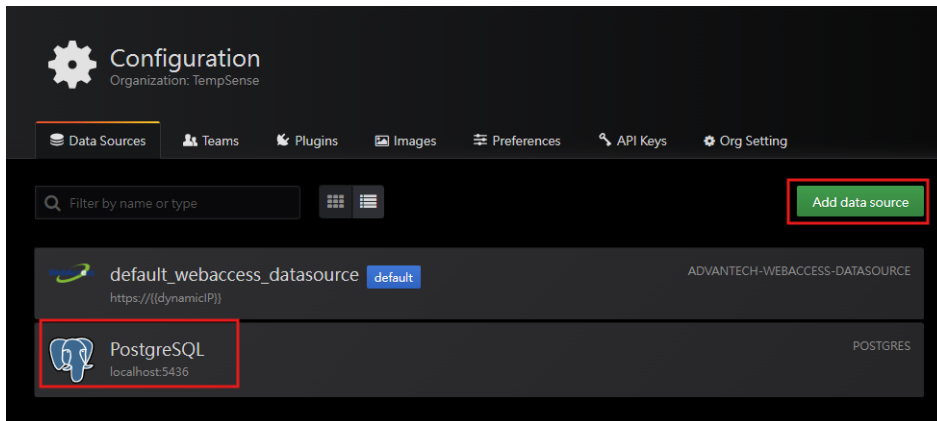


Figure 9.16: Dashboard: Add PostgreSQL data source.

3. Create a **new Dashboard**. Add at least two panels:
 - a **Singlestat** or **Graph** panel bound to your assigned analog sensor tag, and
 - an **Alert-enabled** panel (or an Alert list) showing the current state of your alarm from Activity 2.

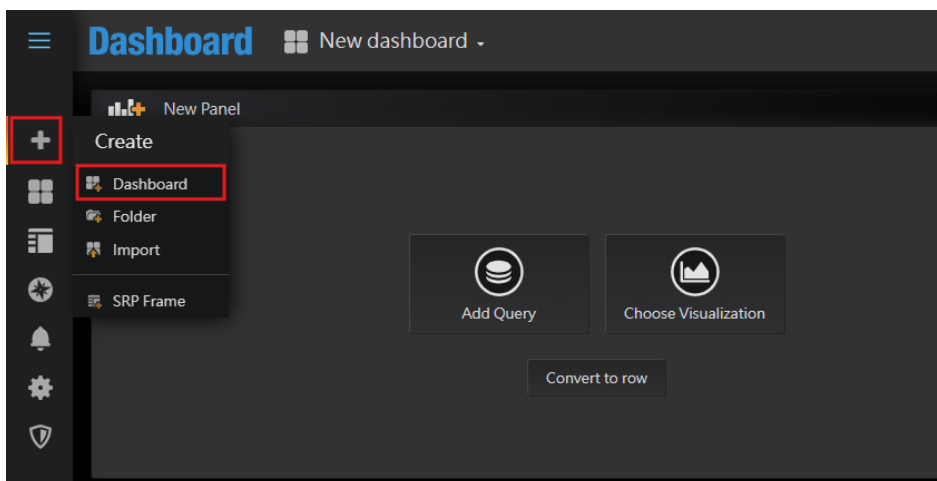


Figure 9.17: Dashboard panel query binding.

4. For each panel, select your PostgreSQL data source and the query/tag that corresponds to your assigned sensor.
5. Label your dashboard clearly with your group number, so it is identifiable in your submitted report.
6. With the machine running, confirm each panel updates live, and that triggering your

alarm condition changes the alert panel's state.

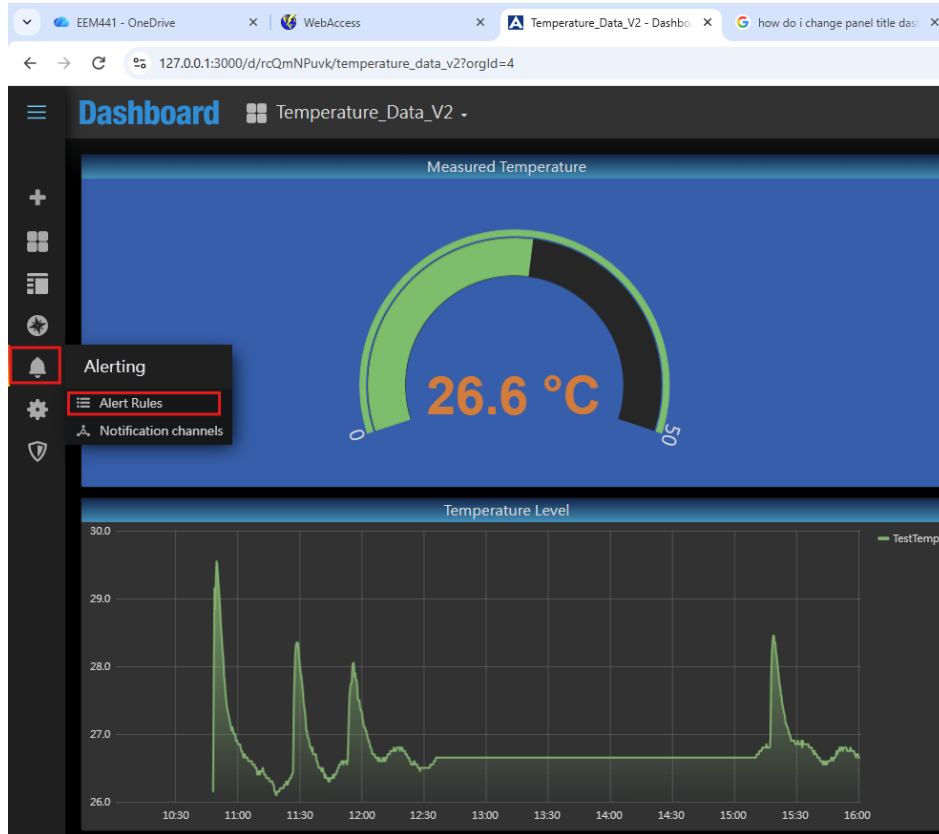


Figure 9.18: Finished dashboard example.

Report

1. Draw the complete data pipeline for your system (PLC → SCADA → database → dashboard), labelling the protocol used at each stage.
2. State your assigned sensor, its PLC DB address, and the WebAccess tag configuration you used to bring it into SCADA.
3. Describe the alarm you configured, including its threshold and priority, and show evidence (e.g. a screenshot of the Alarm Summary) that it triggers correctly.
4. Describe your ODBC bridge configuration (DSN name, database, and the WebAccess ODBC data record source settings), and explain what would go wrong if **Log Data** were left disabled on your tag.
5. Present a screenshot of your finished dashboard, and explain why you chose the panel types you used for your sensor and alarm.
6. Describe one problem you encountered in this experiment (in any layer of the pipeline) and how you diagnosed and resolved it.

Deliverables

1. **System configuration summary:** your PLC, SCADA, and ODBC settings, including the pipeline diagram and tag mapping from the Report questions.

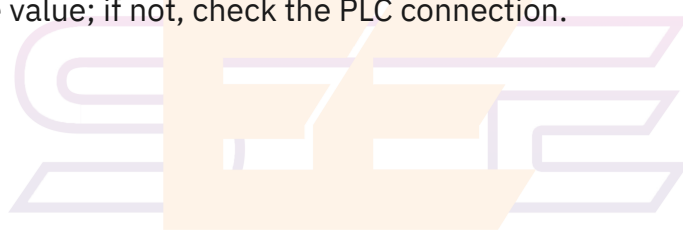
2. **Dashboard screenshot(s):** your finished dashboard, clearly labelled with your group number.
3. **Short demonstration:** a live walkthrough (or short video) showing your dashboard updating in response to the physical sensor, and the alarm triggering correctly.

References

- MTA SkillLab Sdn Bhd, *i-Factory IIoT Training Manual*, Module 2 (IIoT Hardware & Software), Module 3 (Project Creation in SCADA), and Module 4 (Dashboard Viewing and Analysis).
- Advantech WebAccess/SCADA product documentation.
- PostgreSQL ODBC driver (psqldbc): <https://www.postgresql.org/ftp/odbc/versions/msi/>

Appendix: Troubleshooting

This experiment's pipeline has four layers (Table 9.1), and almost every symptom traces back to exactly one of them. Work backward from where the data visibly stops — if the dashboard panel is blank, check whether the database has rows; if not, check whether SCADA shows a live value; if not, check the PLC connection.



Experiment 9

IIoT Monitoring & Alarm Visualization for a PLC-Controlled Machine

Objectives

1. To understand the role of each layer in an IIoT monitoring pipeline: PLC, SCADA, database, and dashboard.
2. To configure communication between a Siemens S7-1200 PLC and Advantech WebAccess/SCADA.
3. To bridge SCADA tag data to a PostgreSQL database via ODBC, so that history is available outside WebAccess.
4. To design a web-based IIoT dashboard displaying real-time sensor status and visual alarms, using the WISE-IoTSuite/Dashboard application.
5. To demonstrate a complete monitoring pipeline, from a physical sensor to a live dashboard panel.

Equipment / Software Required

- MTA SkillLab i-Factory kit: conveyor sorting machine with Siemens SIMATIC S7-1200 CPU 1212C AC/DC/Rly, fitted with vibration, laser-distance, temperature/humidity, and part-counting sensors
- PC/laptop with Advantech WebAccess/SCADA V9.x installed (Project Node and SCADA Node)
- PostgreSQL, with the psqlodbc (x86) ODBC driver and pgAdmin4 installed
- ODBC Data Source Administrator (32-bit) — WebAccess only supports 32-bit ODBC connections
- Ethernet cable and a configured local network connection between the PC and the PLC

Introduction

Industrial IoT (IIoT) monitoring bridges field-level operational technology (OT) — sensors, actuators, and the PLC that controls them — with IT-side visualization and analytics. This experiment builds that bridge one layer at a time, using the same i-Factory conveyor rig and WebAccess/SCADA software introduced in Experiment 5.

Communication protocols used at each layer (Table 9.1): Modbus links the sensors to the PLC; TCP/IP (or OPC UA) links the PLC to WebAccess/SCADA; and ODBC links WebAccess/SCADA to the PostgreSQL database that the Dashboard ultimately reads from.

The full pipeline is shown in Figure 9.1. Each stage depends on the one before it: a tag with no PLC communication shows no data; a tag with communication but no **Log Data** setting never reaches the database; and a database with no rows has nothing for the Dashboard to display. Troubleshooting this experiment usually means working backward through this chain to find where data stops flowing.

Table 9.1: Communication protocols in the IIoT pipeline.

#	Protocol	Role
1	Modbus	Communication between sensor and PLC
2	TCP/IP	Communication between PLC and WebAccess (SCADA)
3	ODBC	Communication between SCADA and the database
4	OPC UA	Alternative standard communication between PLC/equipment and SCADA



Figure 9.1: IIoT monitoring pipeline for this experiment.

Sensor Assignment

Each lab group is assigned one primary sensor/parameter to monitor for this experiment. Working with a distinct analog or discrete signal per group ensures each team independently validates its own signal path (PLC addressing, tag configuration, alarm thresholds, and dashboard binding) rather than inheriting settings tuned for a different sensor type.

Refer to Table 9.2 for your group's assigned sensor, its PLC DB address, and the suggested tag prefix to use in WebAccess/SCADA.

Table 9.2: Sensor assignment by group.

Group	Sensor / Parameter	PLC DB address	Suggested tag prefix
G1	Temperature	DB3, DBD92	G1_Temp
G2	Relative humidity	DB3, DBD88	G2_Humid
G3	Vibration — Z-axis RMS	DB3, DBD76	G3_VibZ
G4	Vibration — X-axis RMS	DB3, DBD72	G4_VibX
G5	Vibration sensor temperature	DB3, DBD80	G5_VibT
G6	Laser distance (analog)	DB3, DBD84	G6_Dist
G7	Good part count	DB3, DBD12	G7_GoodP
G8	Defect part count	DB3, DBD16	G8_DefP
G9	Machine OEE availability index	DB3, DBD24	G9_Avail
G10	Conveyor motor speed	DB3, DBD32	G10_MtrSpd

Every group also shares two common digital signals, regardless of assigned sensor: **Start** (DB1, DBX0.1), **Stop** (DB1, DBX0.3), and **Emergency Stop** (DB1, DBX0.0).

Include at least one of these as a discrete tag alongside your assigned analog sensor, so your dashboard has both a trend and a status indicator to display.

Procedure

Activity 1 — Create the SCADA project and connect to the PLC

Objective: create a WebAccess/SCADA project and bring your assigned sensor in as a live tag.

1. Open a web browser and navigate to 127.0.0.1 (or localhost) to reach the WebAccess/SCADA landing page. Select **Project Management** from the **Platform** dropdown and log in (default account admin, no password).

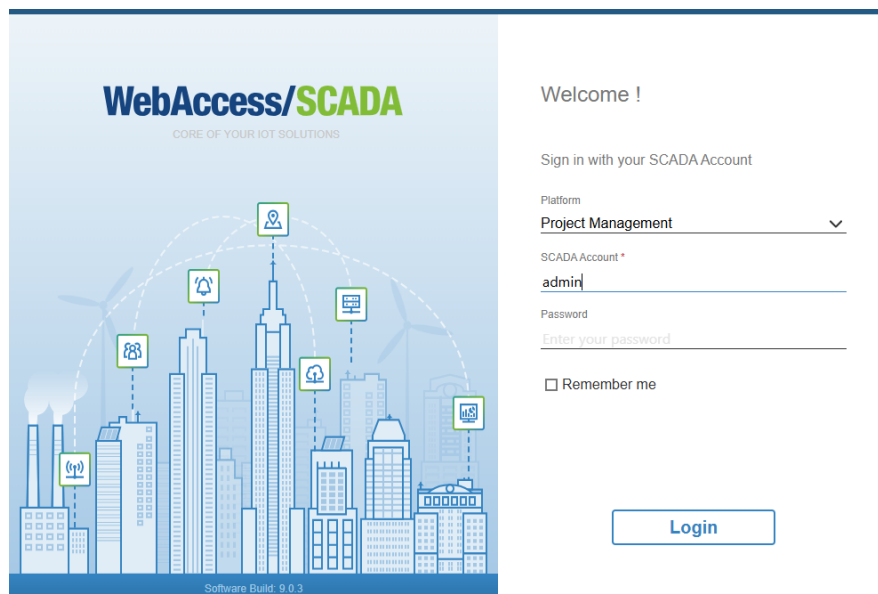


Figure 9.2: WebAccess/SCADA sign-in, with Project Management selected.

2. Create a new project: enter a **Project Name** and **Description**, set the **Project Node IP Address** (127.0.0.1 for local), and keep the default **Primary TCP Port** (4592) and **HTTP Port** (80) unless instructed otherwise.

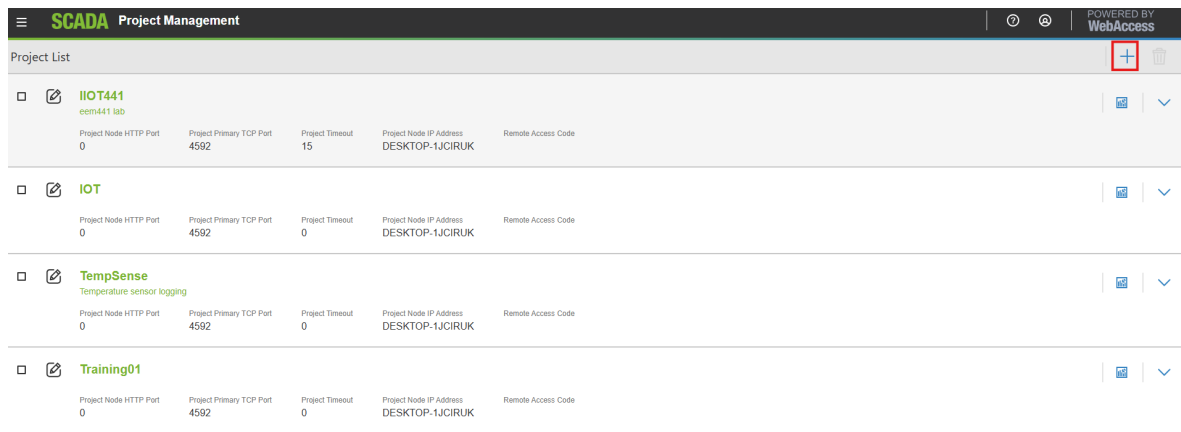


Figure 9.3: Create New Project form.

3. Create a **SCADA Node**: select your Project Node, click **+**, choose **New Node**, and fill in the Primary Node Name, TCP Ports, Node Timeout, and SCADA Node IP Address.



Figure 9.4: Create New SCADA Node form.

4. Add a **Communication Port**: choose **TCP/IP** as the interface, and configure the Scan Time (e.g. 1000 ms) and Auto-Recover Time.

Create New Comport
* Required

Interface Name
TCP/IP

Comport Number *
1

Description
S7-1200 OEEE Monitoring TCP/IP

Scan Time *
1000 ☒ MilliSecond ☐ Second ☐ Minute ☐ Hour

Timeout (Unit:MilliSecond) *
3000

Retry Count *
3

Auto Recover Time (Unit:Seconds) *
5

CANCEL SUBMIT

Figure 9.5: Communication port (Comport) settings.

5. Add a **Device** on that Comport: set Device Type to SiemS7, enter the PLC's IP address and port (102), and set the TSAP in hex for an S7-1200 (01.01).

Create New Device
* Required

Device Name *
Siemens PLC

Description
SIEMENS S7-1200

Unit Number *
0

Device Type
SiemS7

Manually switch Primary/Backup. *
0

TSAP in Hex :Device ID, RackSlot *
03.02

☐ Primary

CANCEL SUBMIT

Figure 9.6: Add Device form (S7-1200).

6. Create an IO Tag for your assigned sensor, using the prefix from Table 9.2 (e.g. G1_Temp), and a second discrete tag for Start, Stop, or Emergency Stop. Set the Address, Data Type, and Conversion Code to match your PLC DB offset.

Create New Tag
* Required

Basic Information

Tag Type
DBD (analog)

Tag Name *
TemperatureSensor

Description
DB DWord Data

Scan Type
Constant Scan

Address *
DBD3,96

Conversion Code
Unsigned Integer

Start Bit *
0

Length *
32

Signal Reverse
☐ Yes ☒ No

Scaling factor 1 *
0

CANCEL SUBMIT

Figure 9.7: Insert IO Tag form (analog sensor example).

7. Start the SCADA Node's Kernel, open **ViewDAQ**, and confirm both tags are updating with live values that reflect what is actually happening on the machine.

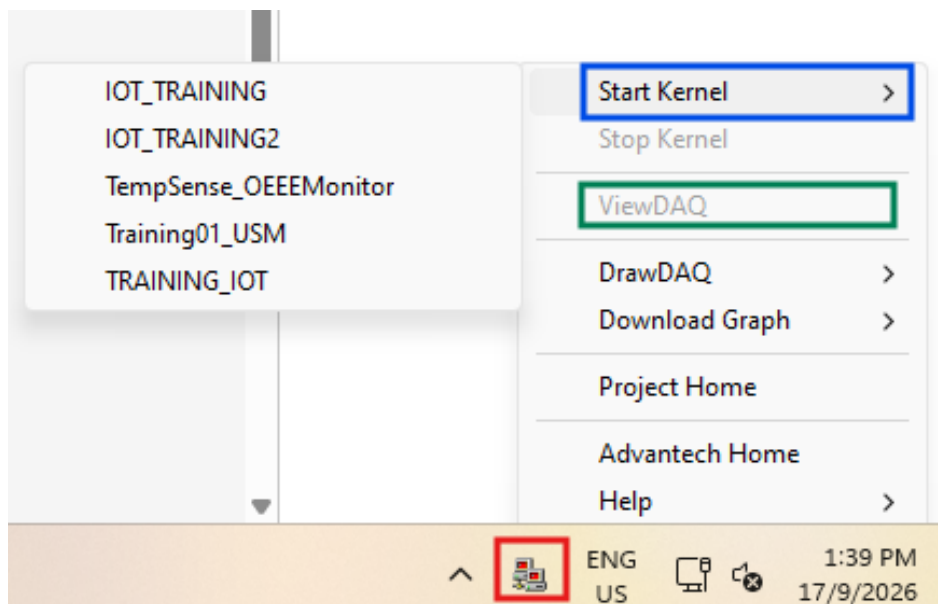


Figure 9.8: Kernel running, viewed in ViewDAQ.

Activity 2 – Configure an alarm

Objective: enable alarming on your assigned sensor, so threshold breaches are visible in both SCADA and, later, the dashboard.

1. Open your analog sensor tag's configuration and enable a non-zero alarm priority (High, High-High, Low, or Low-Low, as appropriate to your sensor).

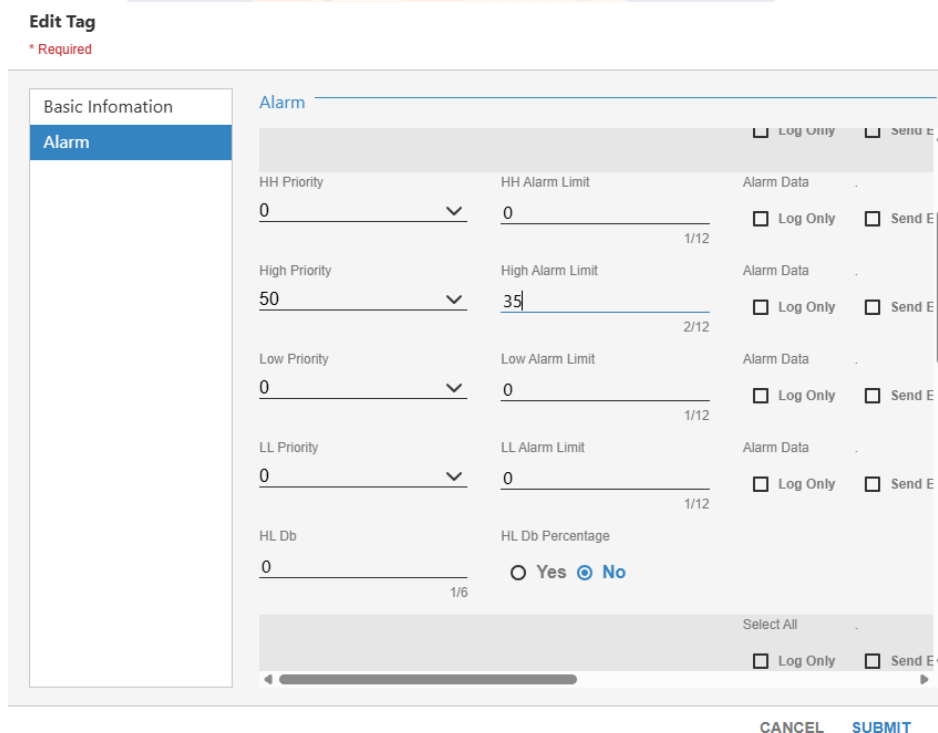


Figure 9.9: Alarm priority configuration.

2. Enable state alarming on your discrete tag (e.g. alarm when Emergency Stop becomes active).

3. In **ViewDAQ**, trigger a manual threshold breach (e.g. briefly block the sensor's path, or press the Emergency Stop) and confirm the alarm appears in the SCADA **Alarm Summary**.

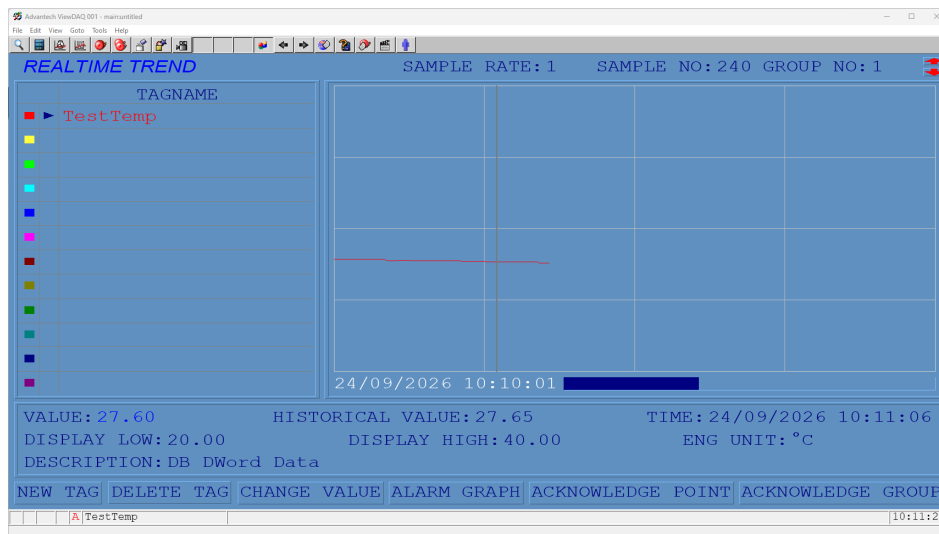


Figure 9.10: Alarm triggered, viewed on the real-time trend.

Activity 3 – Bridge SCADA to PostgreSQL via ODBC

Objective: connect WebAccess to a PostgreSQL database, and enable logging so your tag's history is written there.

1. Confirm with your instructor that PostgreSQL, the psqldbc (x86) driver, and pgAdmin4 are already installed – this experiment assumes they exist; installing them from scratch is outside the scope of this lab.
2. In pgAdmin, create a connection to the PostgreSQL server (Host 127.0.0.1, Port 5436, default credentials provided by your instructor), then create a new database for this experiment.

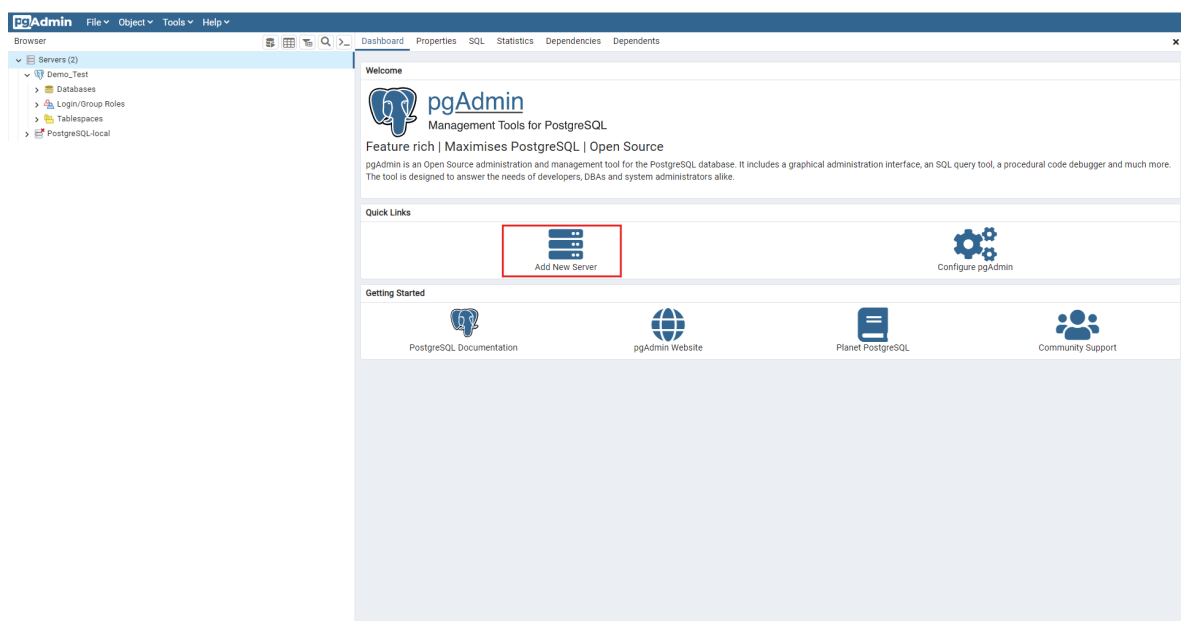


Figure 9.11: pgAdmin: registering the PostgreSQL server.

- Open **ODBC Data Source Administrator (32-bit)**, switch to the **System DSN** tab, click **Add**, and select **PostgreSQL Unicode**. Fill in the Data Source name, Database, Server, Port, Username, and Password to match the database you just created.

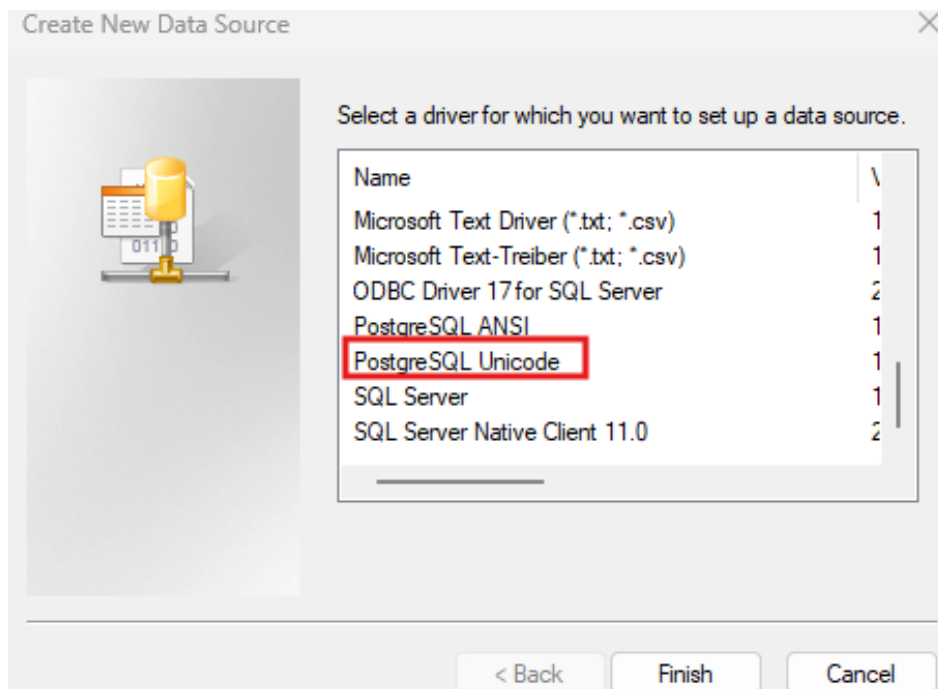


Figure 9.12: ODBC Data Source Administrator (32-bit).

- Back in WebAccess, open the project's **Setting** menu → **ODBC data record source**, and insert a new ODBC entry: a numeric Log Data Source ID, the DSN name from the previous step, and matching login credentials. Test the connection.

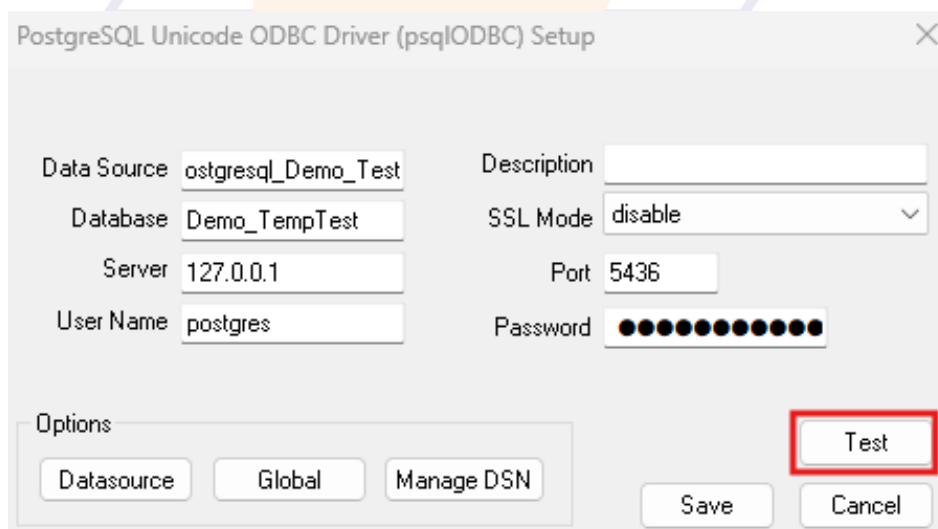


Figure 9.13: WebAccess ODBC data record source.

- On your SCADA Node, enable **Data Log to ODBC (Edit Node → Data Log)**. Then, on each tag you want logged, enable **Log Data** and set the **Data Log Interval** to 0 (record by deadband) to avoid missed samples.

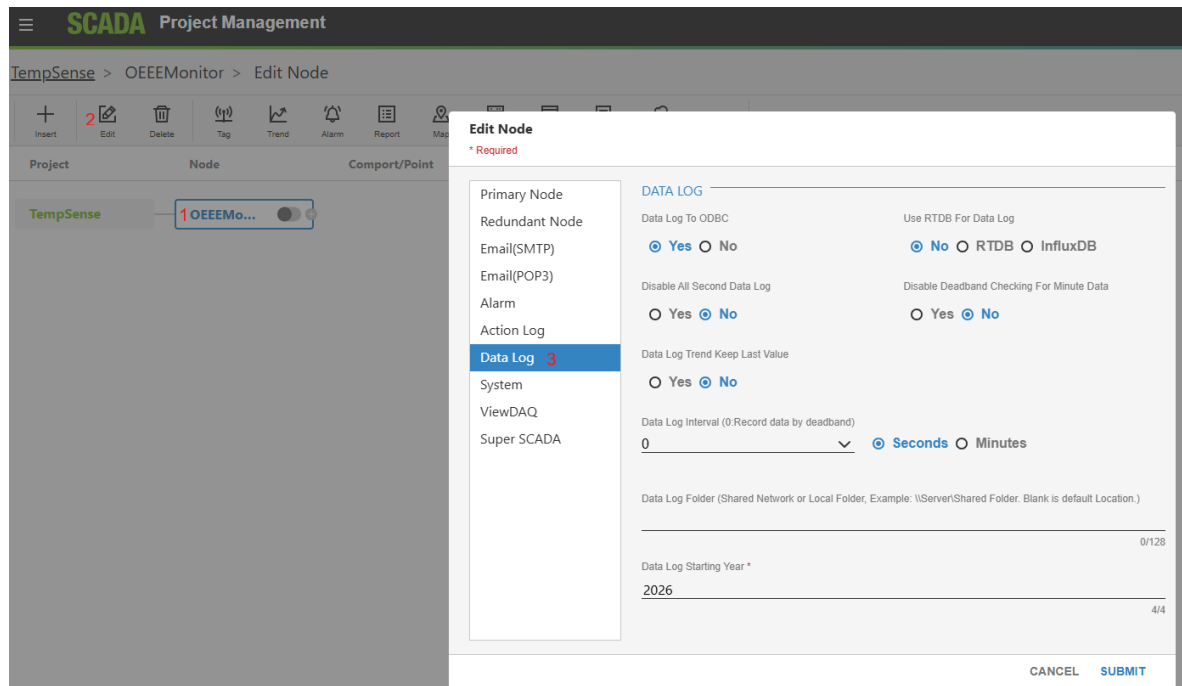


Figure 9.14: SCADA Node Edit Node: Data Log tab.

- Download the project to the SCADA Node, then confirm in pgAdmin that rows are appearing in the corresponding log table (e.g. `bwana_logtable` for analog tags).

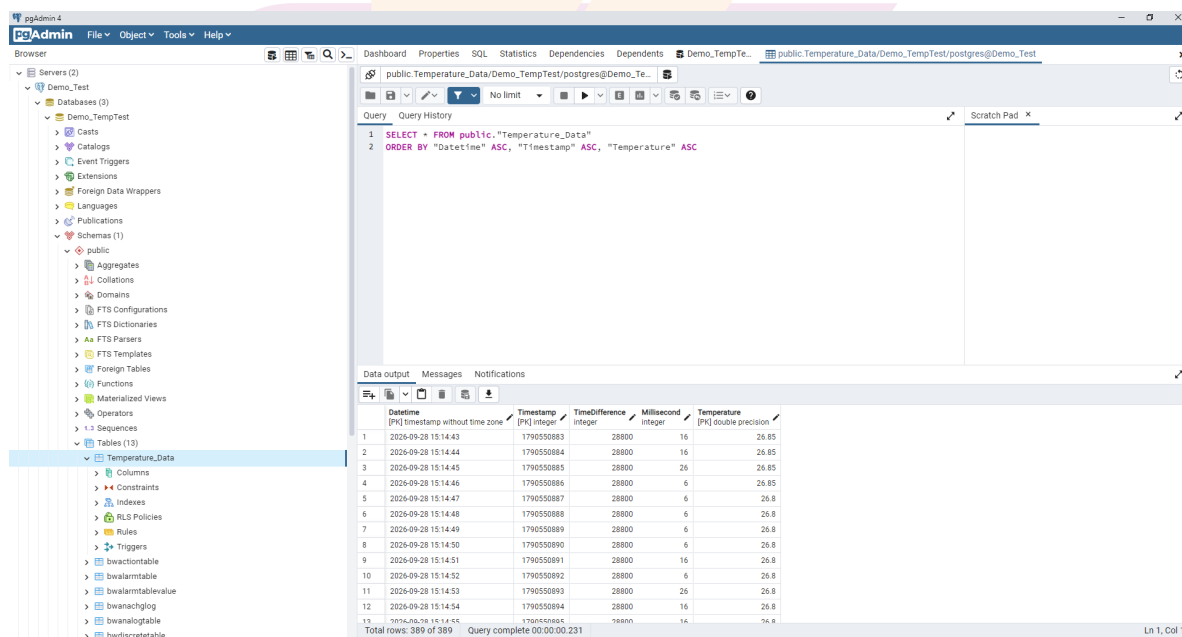


Figure 9.15: pgAdmin: logged rows in the analog log table.

Activity 4 – Build a dashboard

Objective: visualize your assigned sensor and its alarm state on a WISE-IoTSuite/Dashboard panel.

1. From the WebAccess landing page, select **Dashboard** from the **Platform** drop-down instead of Project Management.
2. In the Dashboard app, open **Configuration** → **Data Sources** → **Add data source**, choose **PostgreSQL**, and enter the same connection details used for the ODBC bridge in Activity 3. Click **Save & Test** and confirm a successful connection.

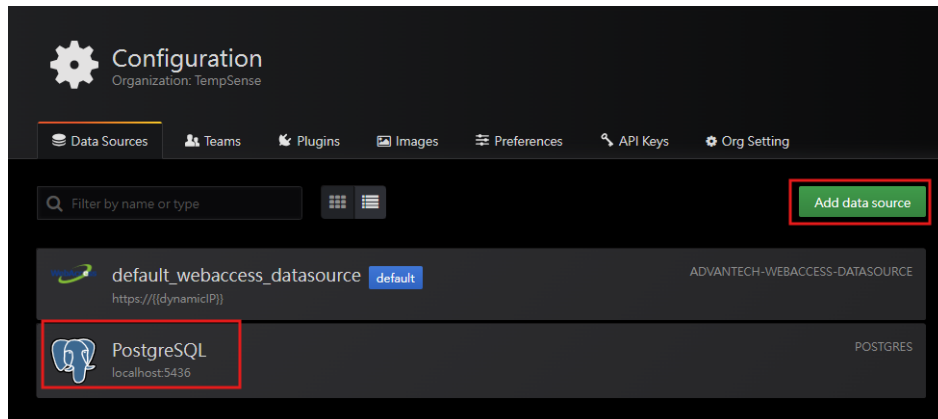


Figure 9.16: Dashboard: Add PostgreSQL data source.

3. Create a **new Dashboard**. Add at least two panels:
 - a **Singlestat** or **Graph** panel bound to your assigned analog sensor tag, and
 - an **Alert-enabled** panel (or an Alert list) showing the current state of your alarm from Activity 2.

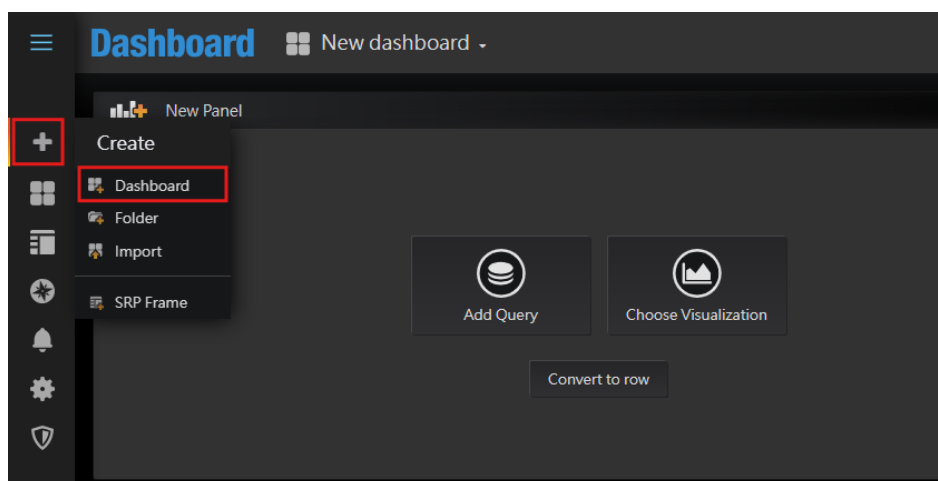


Figure 9.17: Dashboard panel query binding.

4. For each panel, select your PostgreSQL data source and the query/tag that corresponds to your assigned sensor.
5. Label your dashboard clearly with your group number, so it is identifiable in your submitted report.

6. With the machine running, confirm each panel updates live, and that triggering your alarm condition changes the alert panel's state.

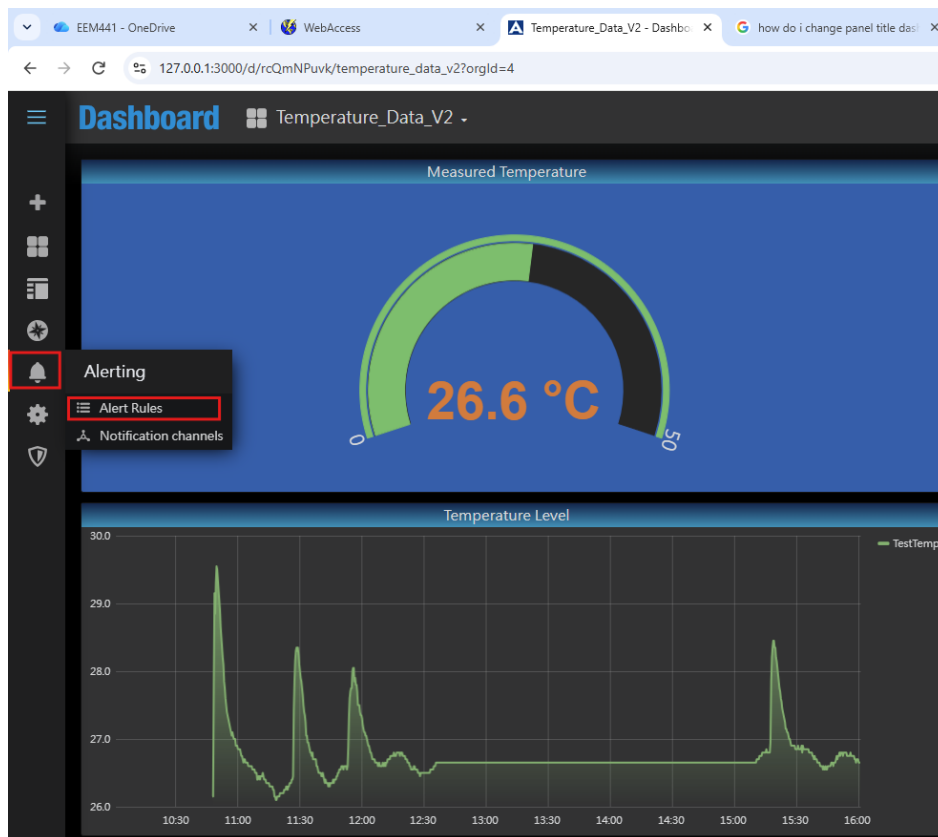


Figure 9.18: Finished dashboard example.

Report

1. Draw the complete data pipeline for your system (PLC → SCADA → database → dashboard), labelling the protocol used at each stage.
2. State your assigned sensor, its PLC DB address, and the WebAccess tag configuration you used to bring it into SCADA.
3. Describe the alarm you configured, including its threshold and priority, and show evidence (e.g. a screenshot of the Alarm Summary) that it triggers correctly.
4. Describe your ODBC bridge configuration (DSN name, database, and the WebAccess ODBC data record source settings), and explain what would go wrong if **Log Data** were left disabled on your tag.
5. Present a screenshot of your finished dashboard, and explain why you chose the panel types you used for your sensor and alarm.
6. Describe one problem you encountered in this experiment (in any layer of the pipeline) and how you diagnosed and resolved it.

Deliverables

1. **System configuration summary:** your PLC, SCADA, and ODBC settings, including the pipeline diagram and tag mapping from the Report questions.
2. **Dashboard screenshot(s):** your finished dashboard, clearly labelled with your group number.
3. **Short demonstration:** a live walkthrough (or short video) showing your dashboard updating in response to the physical sensor, and the alarm triggering correctly.

References

- MTA SkillLab Sdn Bhd, *i-Factory IIoT Training Manual*, Module 2 (IIoT Hardware & Software), Module 3 (Project Creation in SCADA), and Module 4 (Dashboard Viewing and Analysis).
- Advantech WebAccess/SCADA product documentation.
- PostgreSQL ODBC driver (psqldbc): <https://www.postgresql.org/ftp/odbc/versions/msi/>

Appendix: Troubleshooting

This experiment's pipeline has four layers (Table 9.1), and almost every symptom traces back to exactly one of them. Work backward from where the data visibly stops — if the dashboard panel is blank, check whether the database has rows; if not, check whether SCADA shows a live value; if not, check the PLC connection.

Table 9.3: Common symptoms and their likely cause.

Symptom	Likely layer	What to check
Tag shows no value, or a constant 0, in View-DAQ	PLC ↔ SCADA	PLC powered and reachable (ping); Comport/Device IP, port, and TSAP correct; PLC CPU in RUN mode, not STOP
Tag updates in View-DAQ but PostgreSQL table is empty	SCADA ↔ Database	Log Data enabled on the tag; Data Log to ODBC enabled on the SCADA Node; Data Log Interval set to 0
ODBC Save & Test fails (SCADA or Dashboard side)	Database connection	PostgreSQL service running; DSN name, Server, Port (5436), Username, and Password all match exactly
Dashboard panel shows “No data” despite a working data source	Dashboard query	Correct table/tag selected in the panel’s Query settings; time range not excluding all logged rows
Alarm never appears in the Alarm Summary	SCADA alarm configuration	Alarm priority is non-zero; threshold value is actually crossed by the live signal, not just approached

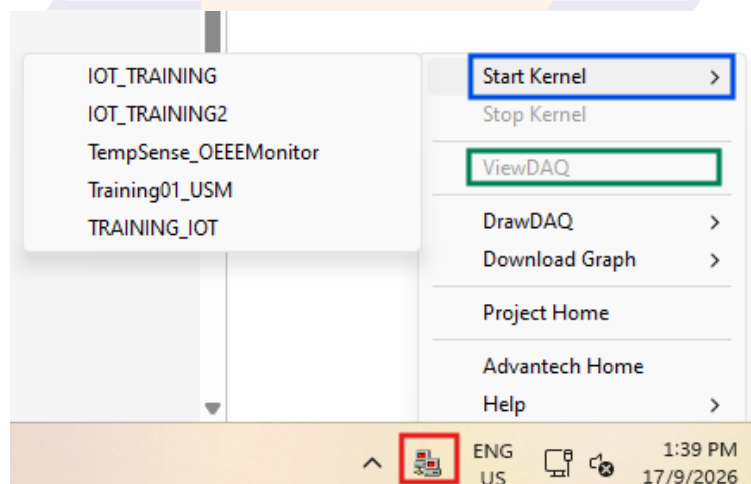


Figure 9.19: Kernel/ViewDAQ status — confirm the Kernel is actually running before suspecting anything further down the pipeline.

	Select	Id	Database Type	Data Source Name (DSN) / IP Address	Description	Connection Test
☐	☐	1	ODBC	brqdata_Access		Connection
☐	☐	2	ODBC	PostgreSQL-local		Connection
☐	☐	3	ODBC	PostgreSQL-local		Connection
☐	☒	4	ODBC	PostgreSQL-local	Logging temperature data	Connection

Figure 9.20: ODBC connection test.

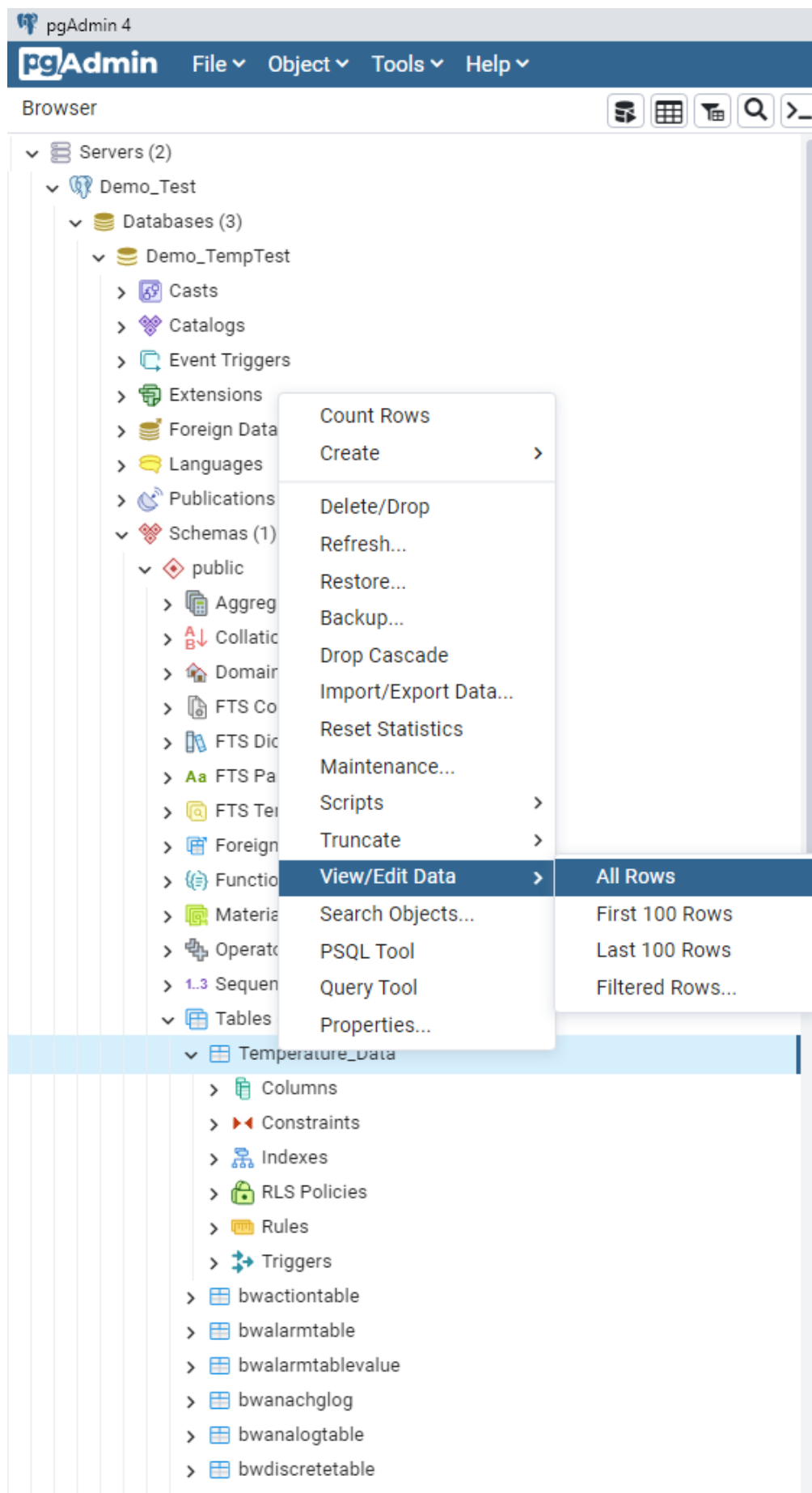


Figure 9.21: Viewing/editing log table data in pgAdmin — an empty result here points back to the SCADA-to-database layer.

Table 9.4: Common symptoms and their likely cause.

Symptom	Likely layer	What to check
Tag shows no value, or a constant 0, in View-DAQ	PLC ↔ SCADA	PLC powered and reachable (ping); Comport/Device IP, port, and TSAP correct; PLC CPU in RUN mode, not STOP
Tag updates in View-DAQ but PostgreSQL table is empty	SCADA ↔ Database	Log Data enabled on the tag; Data Log to ODBC enabled on the SCADA Node; Data Log Interval set to 0
ODBC Save & Test fails (SCADA or Dashboard side)	Database connection	PostgreSQL service running; DSN name, Server, Port (5436), Username, and Password all match exactly
Dashboard panel shows “No data” despite a working data source	Dashboard query	Correct table/tag selected in the panel’s Query settings; time range not excluding all logged rows
Alarm never appears in the Alarm Summary	SCADA alarm configuration	Alarm priority is non-zero; threshold value is actually crossed by the live signal, not just approached

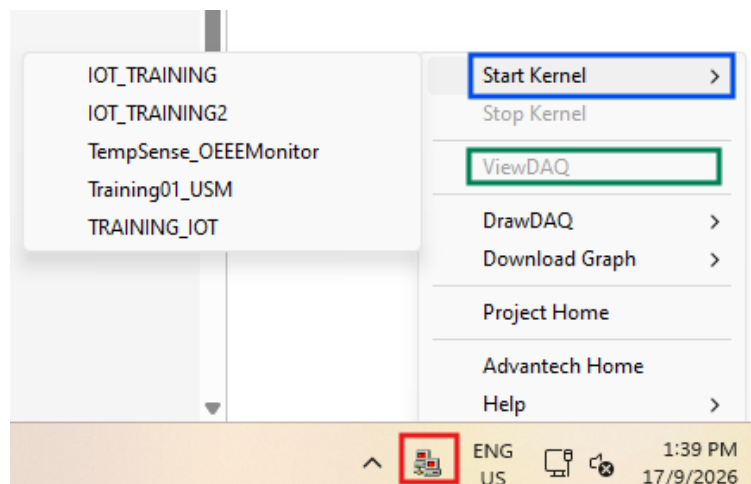


Figure 9.22: Kernel/ViewDAQ status — confirm the Kernel is actually running before suspecting anything further down the pipeline.

	Select	ID	Database Type	Data Source Name (DSN) / IP Address	Description	Connection Test
☐	☐	1	ODBC	Invdata_Access		Connection
☐	☐	2	ODBC	PostgreSQL-local		Connection
☐	☐	3	ODBC	PostgreSQL-local		Connection
☐	☒	4	ODBC	PostgreSQL-local	Logging temperature data	Connection

Figure 9.23: ODBC connection test.



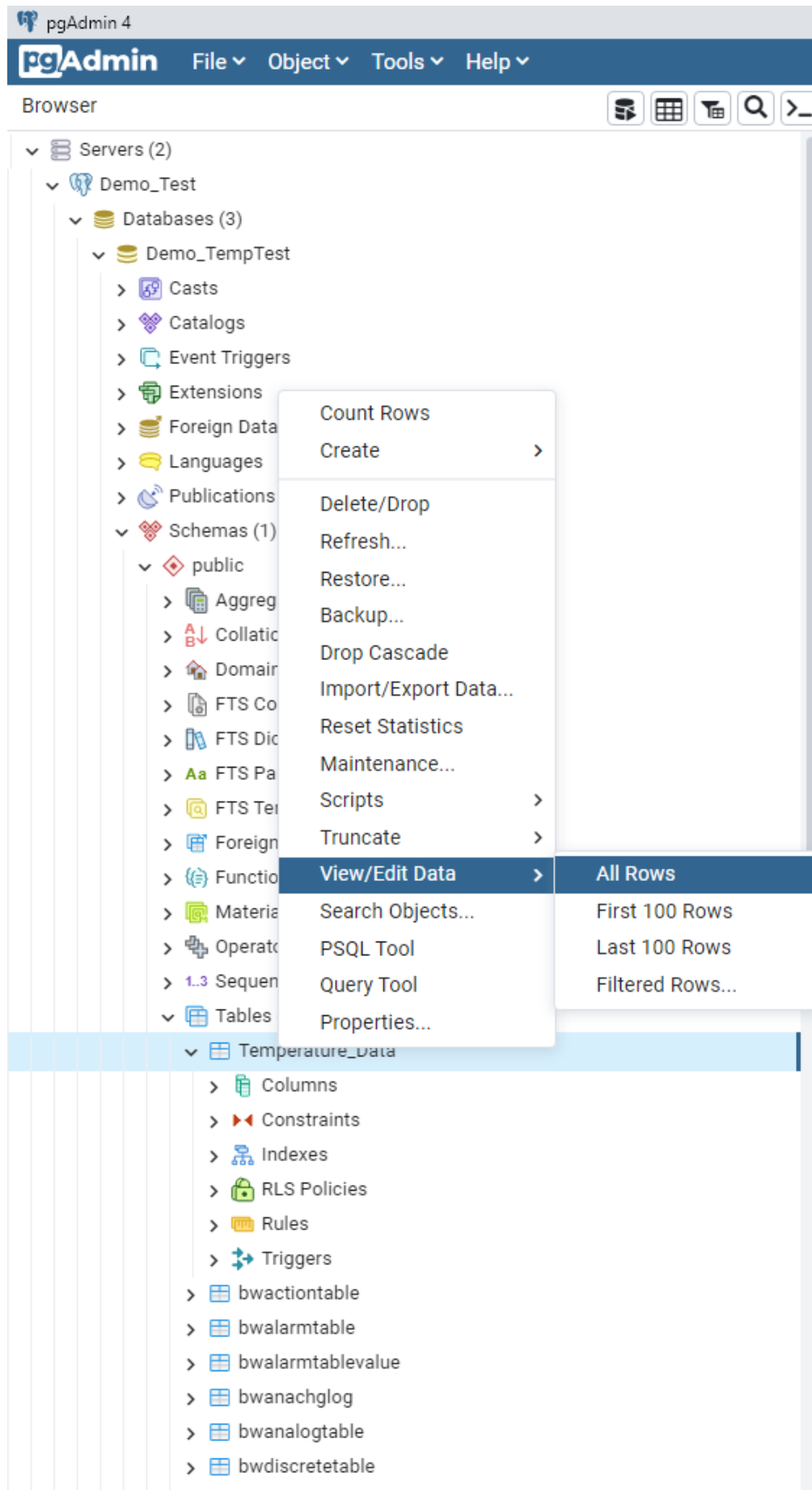


Figure 9.24: Viewing/editing log table data in pgAdmin — an empty result here points back to the SCADA-to-database layer.

